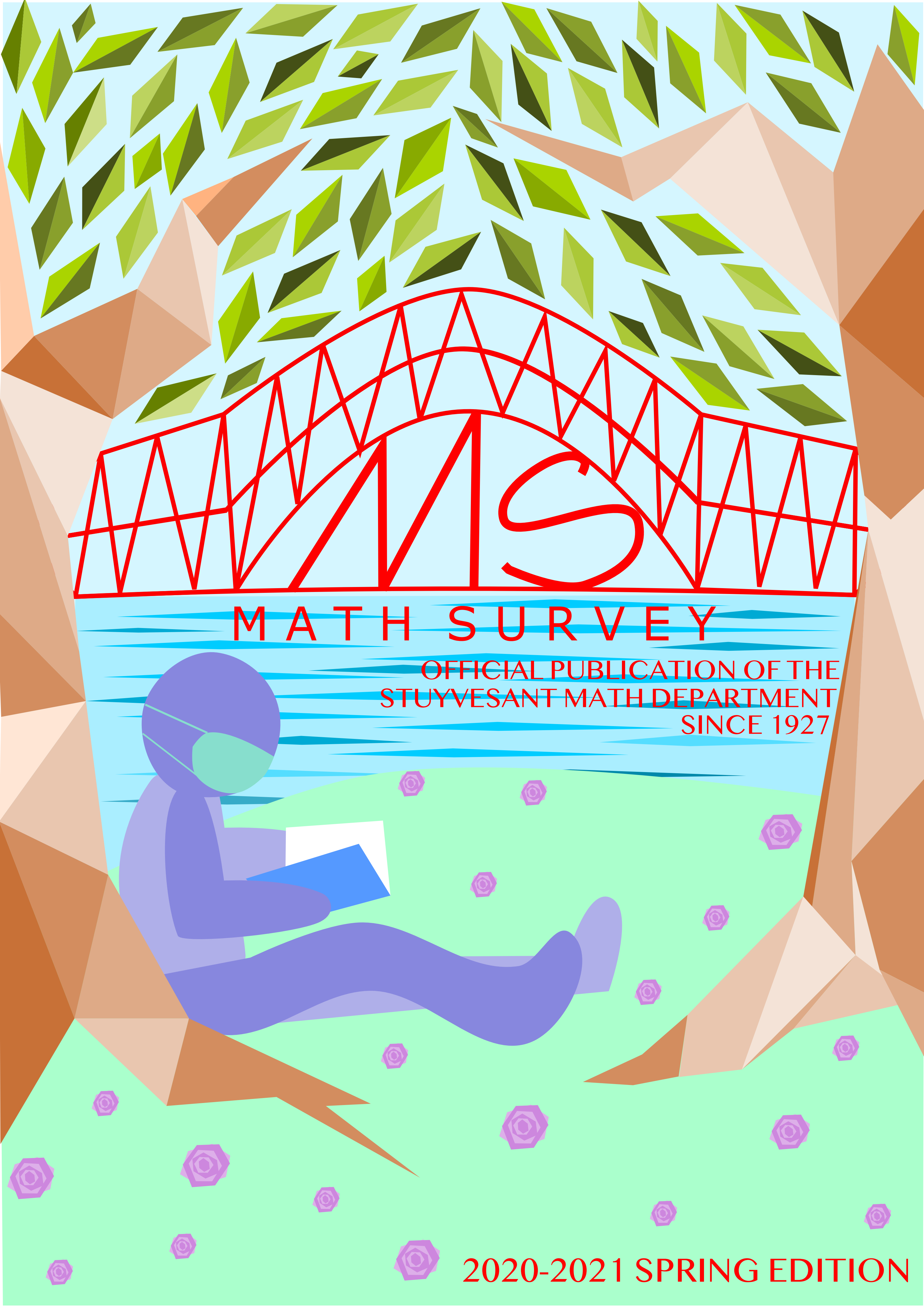
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SINCE 1927
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Stuyvesant Math Survey

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Editor's Note

Josephine Lee

The Olympics are a wonderful metaphor for world cooperation, the kind of international competition that's wholesome and healthy, an interplay between countries that represents the best in all of us.

- John Williams, composer of theme music for the 1984 Summer Olympic Games

Even if you're not athletically inclined, aren't you excited to watch the Summer Games of the 32nd Olympics in Tokyo, Japan? These incredible athletes bring countries together, break records, and tell inspiring stories of overcoming tremendous setbacks to compete on the world's stage. We admire their years of training and experience, perseverance, defiance of limits, dogged pursuit of long-term goals, desire to represent their countries well, and so much more.

We feel the same pride and anticipation and see the same character qualities and habits when we watch or participate in the international olympiads for mathematicians and computer scientists. In the annual International Mathematical Olympiad (IMO) and the International Olympiad in Informatics (IOI), different countries around the world host competitions each year for the highest qualified high school (and sometimes middle school) students in their fields. In Stuyvesant's long history of competition, we've proudly sent some of our best problem-solvers to IMO and IOI, including most recently a 2019 IMO Gold Medalist Milan Haiman competing in Bath, United Kingdom and a 2016 IOI Silver Medalist Calvin Lee competing in Kazan, Russia.

At *Math Survey*, we are proud to continue Stuyvesant's tradition of celebrating achievements, discovery, and progress in problem-solving. We strive to encourage and support math and computer science exploration for all interested students and recognize all kinds of mathematicians and computer scientists, in addition to mathematics and computer science. In each edition, we hope to cultivate the Olympic spirit and potential in any and every student.

We are immensely grateful for everyone who makes this publication possible. To Principal Yu and Math and Computer Science Department Assistant Principal Mr. Eric Smith, thank you for diligently overseeing the continuous growth of the Math and Computer Science Department and for your dedication to the Stuyvesant community. To our faculty advisor Mr. Gary Rubinstein, thank you for all your support that upholds *Math Survey*'s mission. To the extraordinary math and computer science faculty, thank you for your daily work in teaching the next generation of problem-solvers. To the *Math Survey* staff, thank you for fostering the spirit of cooperation and collaboration in math and computer science here at Stuy and around the world.

Foreword

Where Will Math Lead You?
Over 50 Years of David Harbater's
Mathematical Journey and Insight

Professor David Harbater, Class of 1970, is a mathematics professor at the University of Pennsylvania and is most known for his work in Galois theory, algebraic geometry, and arithmetic geometry. While attending Stuyvesant, he was an editor for Math Survey and a co-captain of the Math Team. After graduating from Stuyvesant, he attended Harvard College and graduated summa cum laude in 1974. Harbater then earned a master's degree from Brandeis University followed by a Ph.D. from MIT in 1978, where he wrote a dissertation (Deformation Theory and the Fundamental Group in Algebraic Geometry) under the direction of Michael Artin. In 1995, Harbater was awarded the Cole Prize along with Michel Raynaud for their solution of the long-outstanding Abhyankar conjecture.

This foreword is based on an interview conducted by Josephine Lee. The content has been edited for clarity.

"The beauty of human creativity" at Stuyvesant

I was involved in various math activities at Stuyvesant, with one of them being an editor for *Math Survey*. It was a very different and tedious process since we had to use a typewriter to type the mathematical symbols. With the rise of new technology, people can now concentrate more on the mathematical aspect and less on the technical aspects. We used to go around to classrooms at 10 AM to make announcements saying, "We're selling the *Math Survey*. So if you'd like to buy a copy, I have a whole bunch of copies here." I enjoyed the camaraderie of people staying around after school, typing up, and pasting things down.

We had three co-captains of Math Team: me, Arthur Rothstein, and Julius Collins. We were usually working on one area of math and relating it to other areas of mathematics to have a broad mathematical perspective. We saw how math relates to science and to everything else. It's the beauty of human creativity.

The Hampshire College Summer Studies in Mathematics program, which I learned about through the Math Department, gave me a broader sense of mathematics and a taste of all the math up through college graduation. That was the first time I had the idea of what it meant to be a mathematician after conversing and engaging with professors. It led the way to both college and graduate school. In later summers, I attended the Ross Mathematics Program, first as a participant and again as a counselor.

The Stuyvesant math teacher that influenced me the most was Mark Schwartz. He pushed me to attend NYU for math classes and encouraged me to join the math team and apply to many summer programs. He also advised me to learn how to read math in French and Russian, since mathematics is international.

During my senior year, I studied French for one year, and it was useful when I was traveling to different conferences. It wasn't necessary just to read math, although I do have to read in French all the time because if I'm looking at something written 50 years ago, it might very well be in French.

Communicating and Collaborating

Being a mathematician has a collaborative nature, and it's becoming more collaborative as time goes on because of advancements in technology. When I was getting my Ph.D., most of the articles were published or written by one person, with only some very important articles written by two. And rarely I would see one written by three. These days, most papers have more than one author.

For instance, I am working on finishing one paper, and there are six of us with various perspectives writing from different places. We have grown up on different continents, making it a very diverse group. We communicate with each other well through e-mail and Zoom, which to some extent is solitary. However, after you come up with something and then you tell the others, it bounces back and forth and it leads to much more productivity than you could accomplish otherwise. Individually, a single person could not have achieved it, but together you can.

Although some say that mathematics is mostly done by young people before 30, I've found that's not really true. I love the fact that I was about 40 years old when I won the Cole Prize, and that I've also done significant work since then, including getting published into a very top journal.

Another great aspect of collaborations is that I can work closely with people with both older and younger perspectives: the perspective that includes concepts from 20 years ago and also the other perspective of newcomers to the field thinking about it for the first time who are forming their own ideas and coming to it with fresh eyes.

You usually think mathematics is learned from books, but it's really an oral tradition with an interpersonal aspect. People learn how others think about math, especially their Ph.D. thesis advisors, and that advisor learned that thinking from their advisor, et cetera. You're sort of carrying along this long-term tradition of people thinking in a certain way, so you feel like a part of history. Of course, mathematical thinking evolves over time. But understanding the history often reveals the motivation for that line of thinking.

"Pushing civilization forward" with this "human endeavor"

Mathematicians try to understand phenomena by solving problems and answering questions. Every question has consequences; it's not just a random inquiry. We often think of mathematics as this perfect crystal that exists separate from human existence or even the universe. But mathematics is actually a product of human beings thinking outside our own experience. Mathematical thinking originated from human curiosity, which is an important distinction. Abstract mathematics is filled with all these definitions because there was something they were trying to understand; they were trying to answer a question. So mathematicians should always ask: Why is this concept really important?

When I was a grad student, my Ph.D. thesis advisor taught this to me about math research: You're not just trying to find something that's true or even something that no one ever said before. That's not enough. Here's an example: You write down a 50-digit number. Underneath it, you write another 50-digit number. You draw a horizontal line, and you add them. And the new statement is this number plus this number equals another number. That is a mathematical statement. It's true. If you didn't make any mistakes, no one ever said it before.

But it is not interesting, so it does not count as mathematical research. But what's the mathematical definition of interesting? It needs to deal with people. So that's the weird aspect of mathematics. On the one hand, it has this pure aspect that exists on its own. On the other hand, it's being done by people and people have to find it interesting. It won't make its place on *Access Hollywood*, but it's interesting in some sense of the word.

I think of research as contributing to the long-term process of pushing civilization forward. And to me, that's a high purpose of human beings, to move closer to something greater than themselves. I don't know what that thing is, but good things happen when people create and do things that others appreciate.

Are You Willing to be Wrong? Are You Willing to Fail?

Another professor in my department says that doing mathematics is often just like banging your head against the wall, and you keep banging in hopes that the wall cracks first. When trying to create new mathematics, most ideas that you have don't work. But the ideas that do work will contribute and move you forward a little bit. Then you try to do the next step. And again, you're stuck, and you keep going until you get the next thing that works. So it's a long-term process. You definitely need patience, and you need to be willing to make attempts that don't work. If you're going to do math research, you have to like it. If you don't like it, it's definitely not going to work.

Mathematicians must remember that truth is not just we want it to be. Outside of math, if the truth is inconvenient, they can ignore it. But in mathematics, if you don't have the mindset of "I'm going to find everything that's wrong with my research," someone else will at a later point, maybe even after it's published. You must be willing to work on a problem or conjecture for a long time, and still be willing to have other people tell you that you're wrong and to find that you're wrong yourself.

In coursework, students easily get a misleading idea about mathematics. They hear about all the successes, and they think that everything is known, but in reality, most things are not known. It's just that we don't spend a lot of time in classes telling people, "Oh, we don't know this." So students should be prepared to work on problems which are very difficult. When I was thinking so hard about a certain problem, I felt like my brain was going to explode, but I eventually managed to solve it, and then I got a big prize for it, which was very nice because I felt like I was contributing. On the other hand, I'm still thinking about an even bigger problem that I started working on even earlier. Maybe I'll never solve it, maybe someone else will solve it using my strategy, or maybe they'll combine it with some other strategy and solve it. If my work helps one way or another, that's great.

Learning How to Learn

Too often in math, we don't convey what things are fundamentally about. Textbooks just write down what we know in a very formal way. I tell students they do have to learn math from textbooks, but that's not all. They also have to understand the ideas, the part that's supposed to stay in their heads, and realize that mathematics is not just what's written down. And so I enjoy the challenge of teaching students how to think about what mathematics is really all about and how to use their intuition.

Intuition is crucial. Without it, you won't know how to work with the concepts or what questions to ask when doing research. Although intuition is necessary, it's not sufficient because you also have to know the official stuff that gets written in books.

But if no one is telling you what the sort of "secret part" is, the part that's not written down, then you have to try to come up with it for yourself. What you come up with might be what other people are also thinking of, or it might be something new which might actually be really good. In fact, some of my research emerged from this process: taking things that were written down and asking, "Well, what's really going on?"

The Past, Present, and Future of Math

In the past, mathematics and physics were not distinct fields of study; they were parts of natural philosophy. After they divided, mathematics separated into various branches such as algebra, geometry, etc. But this sub-categorization proved to be counterproductive because mathematicians were building mountains in the center of each of these areas and leaving valleys in between with questions that no one was looking at. Making progress in the middle of a specific area is like taking a rock and carrying it to the very top of the mountain. But with less work, you could move a whole bunch of rocks into a valley and build a really interesting structure, collaborating and merging fields.

What counts as applied mathematics, and where the boundaries lie between different areas, may become even fuzzier as mathematicians and scientists work together more. Very different disciplines might interact with each other in ways that will help each other, creating a richer structure than we have now.

The questions that form the field of mathematics will also change based on what people find interesting, such as abstraction versus not abstraction. For a while, the trend has been going more and more abstract, but in reaction, some mathematicians have moved towards working on more concrete applications. In the future, more contrasting areas of study will be able to coexist.

Mathematics keep changing because it is influenced by what's happening in the world. As mathematicians encounter certain questions they didn't consider before, then they will work on them. For example, I can imagine some area of mathematics coming out of pandemics. Then, that branch of mathematics will take on a life of its own, independent of where it emerged from.

Studying math can lead you to work in all sorts of disciplines - computer science, physics, economics - because it trains your brain and forces you to think very precisely. It lets you focus on how to analyze problems and find structure. Math is more about the structure and method of thinking than it is about numbers.

Many wonder about the extent to which computers rather than humans will be doing math. Computers already work with examples in ways that humans couldn't have before, and in the future, computers will learn more and more by examples rather than definitions. Mathematicians and computer scientists have also been able to write programs to tell the computer how to break proofs down into cases and then handle each case specifically. And then there's even the question of whether computers will be taught, not just how to prove things, but even how to pose interesting questions, which is a complicated process.

Technology will continue to influence mathematics, not just because machines can perform computations, but because technology enables communication and collaboration. Mathematics increasingly welcomes all different kinds of people, even though it didn't before. Smart people who are interested in math can do good in the world, which makes the other aspects comparatively irrelevant.

David Harbater
April 2021

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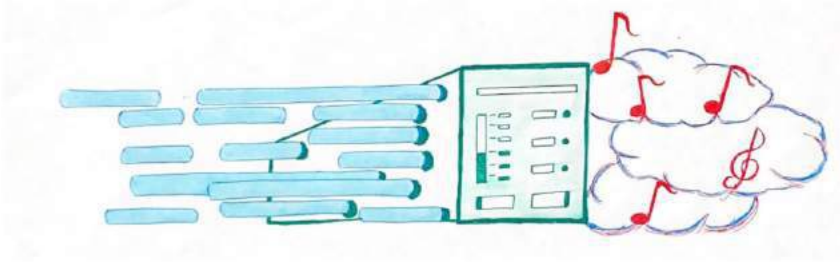
An Overview of the Algorithms Behind the Internet's Most Popular Audio Codec

Victor Veytsman



1 Introduction

This article looks at the algorithms behind the Opus codec, a codec popular because of its versatility and its high performance. We will touch on how analog audio is converted into digital signals, and then how these signals are compressed in various ways to make them suitable for applications such as live audio and video chat and distributing compressed music.

2 What is a Codec?

A codec is an encoder and decoder of a data stream. First, in the hardware world, codecs are devices that convert the analog signal of audio into a digital signal. Then, in the software world, codecs compress the raw data so that it can be stored and transmitted more efficiently. Some software codecs are lossless, meaning they compress the audio without losing any of the raw data, but most are lossy, meaning they lose some of the raw data when compressing it. The challenge in creating a lossy codec is to use as little space as possible while still having high-quality audio.

3 Turning Waves into Bits

The first step in transmitting audio through a computer or storing it in the computer's memory is to turn it into ones and zeros. This is done with **Pulse Code Modulation (PCM)**. Pulse Code Modulation involves three main steps: **Sampling**, **Quantization**, and **Encoding**.

3.1 Sampling

The first step in Pulse Code Modulation is Sampling. When the device receives a wave, it takes samples at regular intervals. Each sample measures the amplitude of the wave.

The **sampling rate** measures how often samples are taken. This number can vary, but usually falls between 40 kHz and 50 kHz (thousand times per second). For example, CD audio is sampled at 44.1 kHz. It is very common for digital equipment to sample at 48 kHz for a variety of reasons that mostly boils down to its ability to deal with the range of human hearing and its many nuances. (This is important because when playing a video, it is convenient when the number of samples played between frames is a whole number.)

3.2 Quantization

The next step is Quantization. In this step, each sample is fit to a range of predefined values, essentially rounding the sample. There are a few algorithms that can select quantization values that are optimal, in the sense that less rounding needs to be done overall. For example, some audio waves are more often close to zero than they are to 1 or -1, so algorithms might use more quantization values in low ranges than in higher ones. This is true of algorithms such as the Mu-law and the A-law algorithms, which are algorithms used to sample audio for digital telecommunication. (Mu-law is used in the United States and Japan, and A-law is used in Europe.) However, it is just as common to chose quantization values that are equidistant from each other. PCM that uses these quantization values is also called Linear Pulse Code Modulation (LPCM).

The **bit depth** is the number of bits each sample uses. A bit is a single one or zero.

3.3 Encoding

Encoding is the final step of Pulse Code Modulation. Exact methods of encoding can differ, but for LPCM it can be as simple as specifying the sample rate and the bit depth at the beginning of the file, and then writing in the data from each sample.

The **bit rate** is a function of the bit depth and the sampling rate, and describes how much data is required to encode a certain length of audio. It is important to know the bit rate because when transmitting audio over the Internet in live instances, one needs to make sure that the bit rate isn't so high that the audio can't keep up with the rate at which the network can transmit data (the bandwidth).

4 What is Opus?

Opus is a codec that has been around since 2010 but is based on earlier technologies in the field. It is a codec that does lossy compression, meaning that while compressing the audio it loses some of the data. It has found widespread usage in applications like Zoom and Discord because of its versatility and its ability to produce great sound in many conditions.

Opus supports sampling rates between 8 and 48 kHz and bit rates between 6 and 510 kB/s (1 kB/s = 8000 ones and zeros per second). It can also seamlessly switch between different sampling rates and bit rates, making it perfect for transmitting over a network that might not be the most stable. It backs this strength up by being fast enough that it can be used for live communication with little delay. Some programs, such as the program "Jamulus," use the codec to allow musicians to play with each other seamlessly over the Internet, despite the fact that musicians can notice a delay of as little as 30 milliseconds.

Opus manages this versatility by operating in three different modes, switching between the modes seamlessly as conditions change:

- The SILK mode, used for speech at lower bit rates
- The CELT mode, used for music and other high-quality recording
- The hybrid mode, in which SILK is used for lower frequencies (up to 8 kHz) while CELT is used for higher frequencies (up to 20 kHz)

5 SILK and Linear Predictive Coding

The SILK mode is based off of the SILK codec, which was first created by Skype for speech. The algorithm behind the SILK codec is **Linear Predictive Coding**, which is also used for speech synthesis.

In Linear Predictive Coding, the audio is encoded in chunks, between 2.5 and 20 milliseconds long. Each chunk includes the first few samples, a prediction coefficient, and a set of errors. The decoder then reconstructs each subsequent sample using the coefficient and the previous samples it has processed. The errors themselves are smaller to transmit than the samples, and their size diminishes as the prediction coefficient becomes more accurate.

When decoding the audio, the decoder uses the following equation:

$$X_t = \sum_{k=1}^{p} a_k X_{t-1} + e_t$$

X_t is the calculated sample, p is the number of samples in the chunk, and e_t is the error for that sample. Most importantly, a_k is the prediction coefficient mentioned earlier. There are several different algorithms that can be used to find prediction coefficients. Opus uses **Burg's Method**, a method originally proposed in the field of geophysics and was first explored for the use of audio in the 1980s. On top of that, Opus includes multiple methods that further increase the accuracy of the coefficients, including methods that analyze pitch and vocal activity.

6 CELT and Modified Discrete Cosine Transform

The other major piece of the Opus puzzle is its CELT mode, which is based off of Xiph.Org Foundation's open codec CELT codec, which would later contribute to the creation of Opus. The main algorithm behind the CELT mode is **Modified Discrete Cosine Transform**.

A **transform** turns a function (such as an audio waveform) into a weighed sum of waves, such that the original form can be calculated by summing each of the transformed waves, each multiplied by their coefficient (their weight). You may have heard of this in the context of Fourier Transforms. In a Fourier Transform, the waves are complex sinusoids, but in Discrete Cosine Transforms, each wave is a cosine function.

When most of the weights are close to zero, the input is easier to store because you can skip storing those weights. This is helpful in the compression of audio and images, and is the reason Discrete Cosine Transforms are used in JPEGs, MP3s, and other file types.

The Modified Discrete Transform Algorithm looks like this:

$$X_k = \sum_{n=0}^{2N-1} x_n \cos\left(\frac{\pi}{N}\left(n + \frac{1}{2} + \frac{N}{2}\right)\left(k + \frac{1}{2}\right)\right)$$

In this equation, X_k is the output frame, x_n is the input, and N is the number of frames in total. What separates this transform from others is that it takes $2N$ data points and produces N outputs, as can be seen in the sum. This is because each input to the transform overlaps with its neighbors, which is helpful in eliminating any artifacts (distortions caused by compression) that come at the beginning and end of each frame.

Opus uses this algorithm in frames, much like the chunks of the SILK model discussed earlier. These frames can be between 2.5 and 20 milliseconds. Once again, Opus includes methods to make the compression more efficient and accurate, including a pre-filter and post-filter. One of their important functions is to separate the gain (the energy, or the volume) from the shape of the sound, to make it easier to transport and restore on the other side. Often, the codec will also opt to use multiple small MDCTs rather than one for the entire frame, in case there is a large burst of energy (a **transient**) in a frame.

7 Conclusion

This article provides a brief overview of the main algorithms that power the Opus codec. In reality, the codec includes much more than what is discussed here. Essentially, the codec is a culmination of decades of research and implementation done by dozens, if not hundreds, of scientists. Its complexity makes it the best codec available for almost all use cases, and with such meticulously thought-out features, it is no wonder that the codec is so widespread across the Internet today.

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Analyzing β-glucuronidase expression in gut microbial populations of MS patients and healthy controls: An intersection of immunology and computer science

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Abstract

Multiple sclerosis (MS) is a chronic and progressive autoimmune disorder of the central nervous system. Studies have observed specific differences in the microbiota of MS patients and some have established a clear link between estrogen variants and MS by observing a gender bias and demonstrating that estrogens have several immunoprotective and anti-inflammatory properties. β-glucuronidase is an exogenous enzyme that deconjugates estrogens into their active form and as such, may contribute to protection against MS. There has not yet been a study that characterizes the interplay between estrogen variants, β-glucuronidase, and the gut microbiome but doing so may help elucidate hormonal and immunological mechanisms underlying MS pathogenesis. We hypothesize that the relative gut microbial β-glucuronidase gene counts and by extension, gene expression, for healthy controls will be greater compared to that for MS patients. To test this hypothesis we assessed the estrobolome to analyze the levels of β-glucuronidase gene expression in control and MS patients. Using data obtained from previous clinical studies, we characterized microbiota populations though the QIIME2 pipeline and performed a predictive metagenomic analysis using PICRUSt2 plugin to generate β-glucuronidase gene counts and allow for pairwise comparison between MS patients and healthy controls. We were unable to confirm our hypothesis that gut microbial β-glucuronidase gene counts for healthy controls are greater than that of MS patients. We were however, able to qualitatively identify several microbial populations that could be correlated with MS pathogenesis which may prove worthwhile to further investigate.

Introduction

Multiple sclerosis (MS) is a chronic and progressive autoimmune disorder of the central nervous system (CNS). Host immune cells recognize and attack the oligodendrocytes, which produce the myelin sheaths necessary to propagate electrical impulses between neurons. T cells that are activated by myelin secrete cytokines, causing chronic inflammation and the blood-brain barrier to become penetrable to other immune cells. Repeated degradation of the myelin sheath leaves behind dysfunctional scar tissues and lesions called plaques that vary in their physiological symptoms depending on their location in the CNS.

Relapsing-remitting MS (RRMS) is by far the most prevalent type of MS, accounting for nearly 85 percent of all cases, and is characterized by bouts of autoimmune attacks that happen months or years apart.[12] There is currently no cure for MS; however, medications like corticosteroids, cyclophosphamide, IV immunoglobulin, and other immunosuppressants can be used to lessen the severity and frequency of relapses.[1] Several risk factors for MS include gender, Vitamin D deficiency, which has been shown to trigger peripheral T cells to cross the blood-brain barrier.[2,3] The human gut microbiota also associated with immune disorders like MS because dysbiosis has been shown to disrupt its normally immunoprotective mechanisms. Consisting of over 100 trillion microorganisms, the gut microbiota harbors many microorganisms that play crucial roles in a large range of physiological functions like digestion and immune activity.

Studies have observed that there are specific microbial taxonomic differences in the gut microbiome of MS patients when compared to healthy controls.[4] Some are associated with the promotion of inflammatory cytokines and overall inflammation. In other studies with experimental allergic encephalomyelitis (EAE), the most widely used animal model to study MS, it has been reported that T regulatory (Treg) cells become increasingly dysfunctional as a result of broad-spectrum antibiotic intervention.[4] This supports the hypothesis that the gut microbial composition may play a large role in MS pathogenesis.

The factor driving the gender bias characteristic of RRMS and its onset overall may be attributed to the influence of gut microbes and gender-based autoimmunity. In pathogen-free female mice, the gender bias of MS pathogenesis can be reversed by transferring gut microbiota from male mice, suggesting an inherent gender-based difference in gut microbiome composition.[5] Moreover, there is clinical evidence that identifies a clear link between estrogen variants and MS.[5] 17β-estradiol (E2) and estriol (E3) are endogenous estrogens produced in humans. E2 inhibits pro-inflammatory cytokines and E3 has been shown to reduce EAE severity in mouse models.[5] A randomized, placebo controlled, double-blind trial conducted with female RRMS patients and E3 found that confirmed relapse rate was lower in the E3-treated RRMS group.[5] As such, lower concentrations of active circulating estrogens may promote a pro-inflammatory breeding ground for MS pathogenesis.

Here, we focus on a subset of the microbiome called the estrobolome which concerns bacteria that are capable of modulating or metabolizing the body's circulating estrogens. Bacteria in the estrobolome regulate circulating estrogen levels by secreting β-glucuronidase, an enzyme that deconjugates estrogen into their active form.[6] Subsequent dysbiosis of the gut microbiome reduces deconjugation and decreases circulating estrogens. This may contribute to the development of MS because estrogen variants, as previously mentioned, have been shown to have several immunoprotective and anti-inflammatory properties.

There has not yet been a study that characterizes the interplay between estrogen variants and the gut microbiome but doing so may help elucidate hormonal and immunological mechanisms underlying MS pathogenesis. To that point, in this study, we assess and characterize the estrobolome in MS and control patients to identify the potential role of β-glucuronidase gene expression on the pathogenesis of MS.

Our hypothesis is that the relative gut microbial β-glucuronidase gene counts and by extension, gene expression, for healthy controls will be greater compared to that for MS patients.

Specific Aims

Our overarching aim is to assess and characterize the estrobolome in RRMS and control patients to identify the impact of β-glucuronidase gene expression on the pathogenesis of MS. We plan to specifically to:

- **Aim 1:** Identify what microbial populations are associated with RRMS.
- **Aim 2:** Identify what microbial populations in the estrobolome express the β-glucuronidase gene by assessing DNA counts in case vs. control.

Methods

Clinical Studies

When choosing studies for viable datasets, we analyzed their study design through their exclusion criteria and methods. The first clinical study recruited 60 MS patients from the Partners MS Center at Brigham and Women's Hospital and 43 controls from the Brigham and Women's Hospital PhenoGenetic project.[7] Their exclusion criteria is comprehensive and as follows: no antibiotic use in the prior six months, no probiotic or corticosteroid use, no history of using immunosuppressive medications, no history of gastroenteritis or autoimmune disease, and no travel outside of the country in the prior month. None of the MS patients had an active relapse at the time of sampling, and pregnant women were excluded. Some elements of the sample population to note include that the control population consisted of proportionally more females than males (86 percent to 14 percent) whereas the gender disparity for the MS population, although still apparent, was significantly reduced (68 percent to 32 percent). All subjects were Caucasian except for two Black and one Hispanic MS patients. For sample collection, stool samples were collected from the subjects in stool collection containers at any time of day without dietary restrictions. DNA was extracted from the stool samples using the PowerSoil DNA Isolation Kit. To assess only bacterial DNA, they amplified bacterial DNA with a specific probe for 16s rRNA. Finally, in order to assess the microbial community structure of the samples, they generated DNA libraries for 16s rRNA using the V3-V5 regions. This enabled them to later amplify and sequence the V4 regions with Next Generation Sequencing and thus identify which microbial populations were present in the samples.

The second clinical study recruited 45 patients with MS, none of whom had an active relapse at the time of study, and 44 healthy controls.[9] Their exclusion criteria is as follows: no patients suffering from infectious diseases and no antibiotic treatment during the collection of fecal samples. From these patients, they obtained 16s rRNA and metagenomic sequences from fecal samples which were then transported at 4 degrees C to the laboratory in a plastic bag containing a disposable oxygen-absorbing and carbon dioxide-generating agent so anaerobes sensitive to oxygen would survive the trip. Once in the lab, the fecal samples were suspended in phosphate-buffered saline containing 20 percent glycerol, immediately frozen using liquid nitrogen and stored at -80 degrees C until use. The bacterial DNA was isolated and purified using enzymatic lysis methods, presumably similar to the PowerSoil DNA Isolation Kit used in the first clinical study. They sequenced the V1-V2 regions of the 16s rRNA gene and amplified them by PCR. They then used whole genome sequencing analysis with IonProton platforms to obtain metagenomic sequences.

Table 1. Comparison of the Methods of the Clinical Studies

Title of publication	Alterations of the human gut microbiome in multiple sclerosis (Jangi et al)[7]	Gut microbiome of treatment-naive MS patients of different ethnicities early in disease course (Ventura et al)[9]
Experimental Design and Exclusion Criteria	Untreated patients were treatment naïve or received no steroid treatment in the previous month. Subjects were to have had no other treatments in the past six months including antibiotic or probiotic use and corticosteroids. They were also required to have no history of gastroenteritis, irritable bowel syndrome, bowel surgery, inflammatory bowel disease or other autoimmune diseases where immunosuppressive medications were taken. Subjects could not be pregnant nor could they have travelled to outside countries in prior months. None of the MS patients had active relapse, and treated patients received beta-interferon or glatiramer acetate for at least six months leading up to the study.	Subjects of mixed races were not included and all subjects were between the ages of 18 and 70. The only allowed medications were NSAIDs and analgesics and all subjects were to have had no prior or current use of disease modifying or biologic therapies for BRX, recent use of antibiotic therapy, extreme diet, inflammatory bowel disease, or prior GI tract surgery which all may have left permanent residues on their microbiome.
Number of Subjects	60 MS patients (57 Caucasian, 1 Hispanic and 2 African American) and 43 healthy controls. In the context of our study, we only analyzed the untreated MS patients (n=27), excluding patients undergoing beta-interferon or glatiramer acetate treatment.	45 MS subjects (15 Caucasian, 16 Hispanic and 14 African American) and 44 healthy controls
Sequencing Method	16s Sequencing	Whole Genome Sequencing
Sample Type	Stool	Stool
Notable Points	MS cohort had more males	Three stool samples from African Americans were collected more than six months after diagnosis.
	Most of the subjects were Caucasian; there were only two Black MS patients and one Hispanic MS patient	

Data Analysis

We used the QIIME2 pipeline to identify microbial populations present in the datasets retrieved from the two clinical studies, which contained raw 16s rRNA sequences. We imported the sample data and screened them using DADA2 for demultiplexing and denoising before utilizing the pipeline to align the resulting sequences with the GreenGene database. This enabled us to quantify core metrics like alpha and beta diversity while also developing a phylogenetic tree to identify observed OTUs as well as their differential abundance. For alpha diversity, we considered the Faith, Shannon and Observed Features metrics to a sequencing depth of 50,000 base pairs. For beta diversity, the UniFrac method we used is a good fit because it measures the phylogenetic distances between sets of taxa based on mutual lineages while taking into account degrees of similarity between 16s rRNA sequences.

With the alignment data, we then used the PICRUSt plugin in QIIME2 to perform a predictive analysis on the raw 16s rRNA sequences. Both of these analyses generate β-glucuronidase gene counts specific to each microbial population present for pairwise comparison between MS patients and healthy controls.

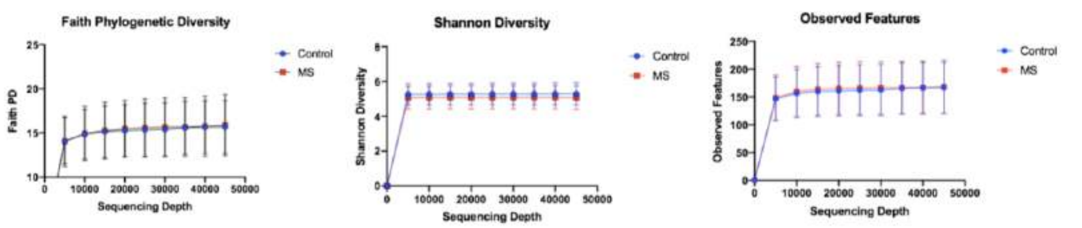
Results

We visualized our results using QIIME2 View, but we re-visualized several figures using GraphPad Prism.

Across all three alpha diversity metrics, there was no significant difference in alpha diversity between MS patients and healthy controls regardless of the sequencing depth. This trend was also observed in the clinical study we retrieved the data from. This suggests that similar microbial populations are present in the gut microbiomes of MS patients and healthy controls.

To assess differences in community structure (beta diversity) we used the emperor PCoA plots generated with QIIME2 to identify the differential microbial diversity between MS patients and controls. Our PCoA for weighted data has two clusters of healthy controls and MS patients (Figure 1b). Weighted data takes into account phylogenetic diversity by mapping the sequences onto a tree for comparison whereas unweighted data looks at the sequences at face value. We then performed unpaired, nonparametric Kolmogov-Smirnov tests on the data, obtaining an insignificant p-value for unweighted data but a p-value of 0.003 for weighted data.

Alpha and beta diversity analyses are helpful because alpha diversity is inversely related to dysbiosis and MS pathogenesis while beta diversity is helpful to distinguish between separate microbial communities and characterize differences in the microbiota of MS and control patients.



Next, we assessed the microbial composition of each subject to identify taxonomic groups present in each subject and their differential abundance. Each column represents a subject, and each color represents a specific OTU. The wider each column is, the more abundant the taxa is.

We then identified OTUs that were more abundant in one group than the other for further analysis regarding how they may have an effect on MS pathogenesis (Figure 2). *Akkermansia muciniphila* and Bifidobacterium adolescentis were abundant in untreated MS cases while the other OTUs were abundant in healthy controls.

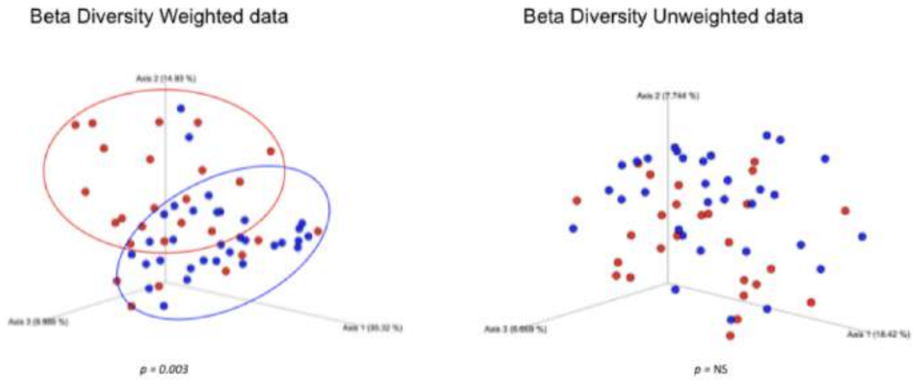


Figure 1: a) Comparison of alpha diversity metrics between MS and control patients. b) Difference in phylogenetic distances between MS patients and controls regarding community structure.

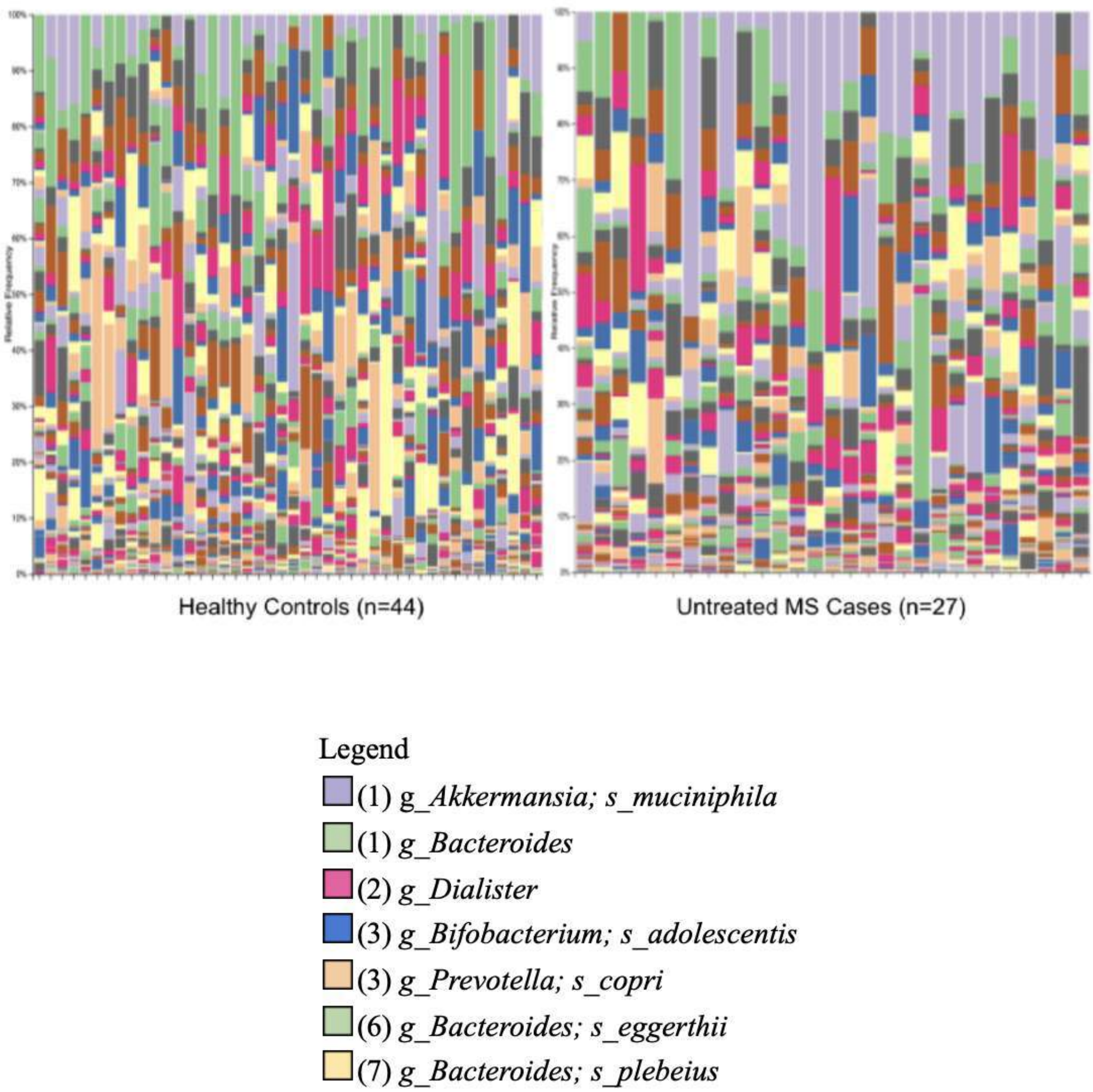


Figure 2: Microbial species present and their differential abundance in healthy controls and untreated MS cases

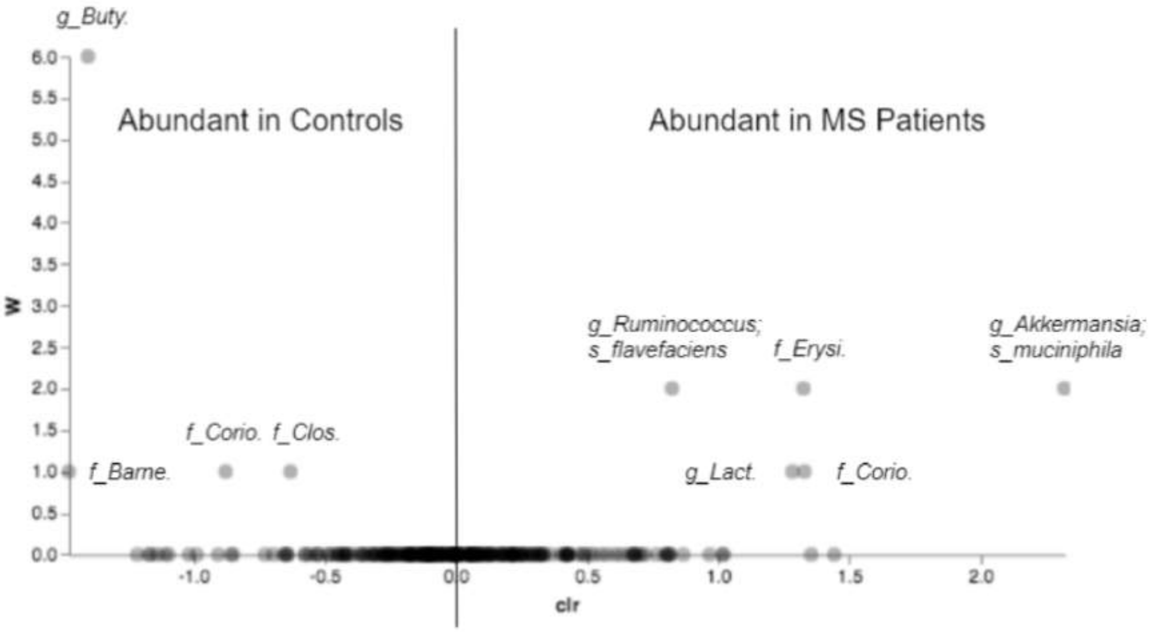


Figure 3: L7 Volcano Plot of the differential abundance of OTUs present in MS and control subjects

Additionally, we quantitatively assessed the differential abundance of OTUs present in the 71 subjects, identifying specific OTUs that were more abundant in either MS patients or healthy controls. The y axis (W) refers to absolute abundance. The centered log ratio (clr) normalizes high composition data sets and in our study, a positive ratio indicates the associated OTU is significantly more abundant in MS patients whereas a negative clr indicates the opposite, that the OTU is significantly more abundant in healthy controls. Notable OTUs abundant in healthy controls include microorganisms in the family Butyricimonas and Barnesiellaceae and notable OTUs abundant in MS patients include Akkermansia muciniphila and *Ruminococcus flavefaciens*.

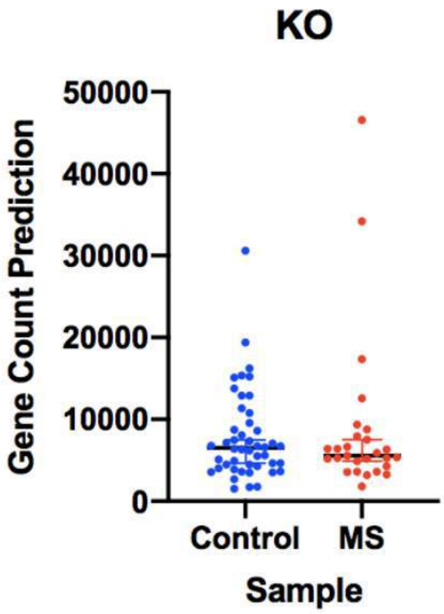


Figure 4: β-glucuronidase gene count metagenome prediction comparison between control and MS patients

Finally, we assessed β-glucuronidase gene count through PICRUST2 metagenome prediction analysis and performed an unpaired, nonparametric Kolmogov-Smirnov test on our gene counts and obtained an insignificant p-value. However, it does appear that the median gene count for controls is higher than that for MS patients.

It is important to note too that these predictive analyses based on differential abundance testing can vary between amplicon-based metagenome predictions like in our case and those utilizing shotgun metagenomics sequencing data.

Discussion

In this study, we explored the estrobolome to characterize the estrobolome in RRMS and control patients and the impact of the expression of the ß-glucuronidase gene on MS pathogenesis. We identified microbial populations associated with RRMS and those that express the β-glucuronidase gene by performing a pairwise analysis of β-glucuronidase gene count between healthy controls and MS patients. This was done using datasets from two previous clinical studies and by taking advantage of bioinformatics platforms like QIIME2 to assess differences in microbial communities and structure and other state-of-the-art pipelines like PICRUSt for predictive metagenomic analysis of the ß-glucuronidase gene. Because β-glucuronidase deconjugates estrogens, which have been shown to have anti-inflammatory and immunoprotective properties, we hypothesized that the relative gut microbial β-glucuronidase gene counts and by extension, gene expression, for healthy controls will be greater compared to that for MS patients.

We first assessed the alpha diversity of all of our samples obtained from the two clinical studies, one dataset generated from 16s sequencing and the other from whole genome sequencing, using the Faith, Shannon, and Observed Features metrics up to a sequencing depth of 50,000 and were able to confirm observations made by other studies that similar microbial populations are present in the gut microbiomes of MS patients and healthy controls. There was a significant phylogenetic difference in microbial community structure between the microbiome of MS patients compared with that of healthy controls (Figure 1b), $p < 0.05$.

Since there were differences in microbial community structure, we then assessed the microbial composition of each subject to identify specific taxonomic groups present in each subject and their differential abundance. With this and a further analysis on the differential abundance of these OTUs, we were able to qualitatively and quantitatively identify several microbial populations that were more abundant in either MS patients or healthy controls.

Additionally, we were not able to confirm our hypothesis that gut microbial β-glucuronidase gene counts for healthy controls are greater compared to that for RRMS patients because the p-value for the t-tests for our gene counts between the two groups was not significant. This suggests that β-glucuronidase gene expression may not be as directly correlated to MS pathogenesis as we had predicted.

A caveat of our analyses, especially our gene count predictive analysis, is that we were limited by the computing power of our personal computers and bottlenecked by downloading large files for other analyses. Because of this computational capacity and limited time, we were unable to perform whole genome analysis of the alignment data calibrated to population abundance using MetaPhlAn and HUMAnN on the confirmed microbial populations. A potential future direction for this study would be to secure increased computing power in order to consider more subjects during predictive analyses, analyze the whole genome data sets, and further analyze the impact of selected OTUs. It may prove worthwhile to study these microbial populations in greater detail to better understand their physiological significance in the human body, particularly in the estrobolome, as they may be correlated with MS pathogenesis or confer protection against MS.

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The Bottom Line: Musings on Linearity in Math and in Life

Xiaoshen Ma

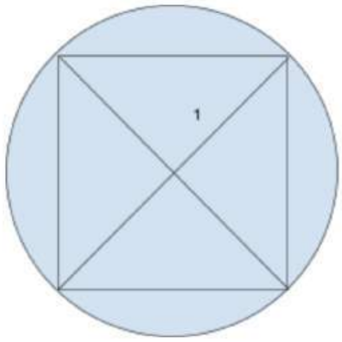
1 Introduction

In early April, I sent my math teacher a tweet on behalf of my friend Maxwell that said, "i used to think the intermediate value theorem was overrated; now I think it's underrated. thus there must have been a point in time when." My math teacher responded with, "jokes on him. He's assuming thinking is continuous and I've seen plenty of contradictions to that." After a few years of teaching grubby high school kids, I'm sure he's seen all kinds of things. At the end of the day, what does math have to do with thinking? If we represent everything with a line, do we also use hearts, squares and triangles? My friend Alyssa on the soccer team probably thinks in truncated icosahedrons when she's worshipping Marcus Rashford. Though I don't have an answer to any of the questions mentioned above, I do have some examples of how these pesky lines wiggle into our daily lives.

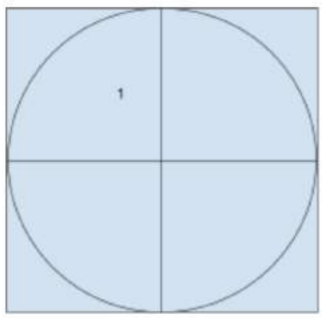
2 Linear Reasoning

Do you support standardized tests? Well, you probably also want a wall on the Mexico-United States border. Not taking any sides here, but doesn't this sound like the last Reddit fight you got into? This line of thinking is linear, though not very continuous. Linear reasoning is a systemic style of thinking, where one idea proceeds another like points on a line. It all goes back to fifth century BCE Greece when people were working with whole numbers and messing around with the Pythagorean Theorem. Someone from the Pythagoras school realized that no isosceles right triangle can be represented using three whole number length sides. Therefore, some believed that the hypotenuse of isosceles right triangles and π were simply not numbers. This was pretty mind-boggling, and legends say that Hippasus, the person who proved the square root of 2 is irrational, was drowned in the sea by his colleagues for suggesting the theorem. Luckily, the idea stuck around. Mathematicians like Euclid and Archimedes later understood that measuring with only whole numbers is a thing of the past and were ready to move on. If circles can't be measured, then how have people been making wheels?

It doesn't seem very continuous for me to go from whole numbers to calculus, but what better way to find the area of a circle than to use limits? The method of exhaustion finds the area of shapes, such as circles, by inscribing a sequence of polygons inside of it. The areas of these polygons then converge to the area of the shape. It seems like the idea has been popping up from fifth century BC's Antiphon, then from Eudoxus of Cnidus, and then from Liu Hui in China. Euclid uses the method to prove six propositions in book 12 of the *Elements*, and Archimedes used it to measure the circle. Starting from an inscribed square, we get a pretty easy but bad estimate. If the circle's radius is 1 unit, then the square's area would be 2 units^2.



Now let's consider a circumscribing square! It's also a pretty bad estimate. With a radius of 1 unit, this square would have an area of 4 units^2. Knowing that our circle's area is π, we just narrowed down the value of π to between 2 and 4!



What? Not very impressive? We at least know that π is a number and not some unmeasurable wheel! Archimedes did go further than squares though. He bounded π with the numbers $\frac{223}{71}$ and $\frac{210}{70}$ by inscribing and circumscribing 96-gons about the circle. Some of his other findings using the method of exhaustion include:

1. The area bounded by the intersection between a line and a parabola is $\frac{4}{3}$ of the area of a triangle that has the same base and height.

2. The areas of ellipses are in proportion to rectangles with sides that have the same measure as the ellipsis' major axes and minor axes.
3. The volume of a sphere is 4 times the volume of a cone that has a base with the same radius of the sphere and a height equal to the radius of the sphere.
4. The volume of a cylinder with a height that is equal to the diameter of its base is $\frac{3}{2}$ of the volume of a sphere with the same diameter.
5. The area bounded by a spiral rotation and a line is $\frac{1}{3}$ the area of a circle with a radius that has the same measure as the line segment.

The first successful evaluations of infinite series also used the method of exhaustion.

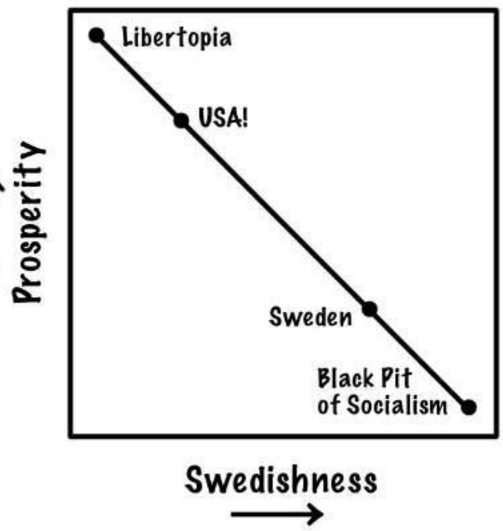
You may be wondering how any of this has to do with linear reasoning, but we're almost there. First off, meet Greg.

Throughout his childhood, his favorite compliment is that he's "well-rounded." Why may a circle be proud of being round? you ask. Well, that's because Greg fooled them all! He's actually a 325325325325325325-gon! Did the mysterious smile give it away? In calculus, you calculate many things by breaking them up into small lines, or the slope at one point. The more you zoom in to a curve, the straighter it becomes. And this is true for all curves!

Well, this is where linear reasoning comes back into the conversation. Thinking in a systematic way, where one point connects to the next with a straight line seems fair, so why do I still feel funny inside when Reddit_b0i_THEb0sser accused me of being a sus-scrofa-domesticus when I said I just want to eat food and sleep this summer? The reason may be that everything in life does not always act as lines! We just like to think in lines because on a super small scale, curves are very similar to lines.

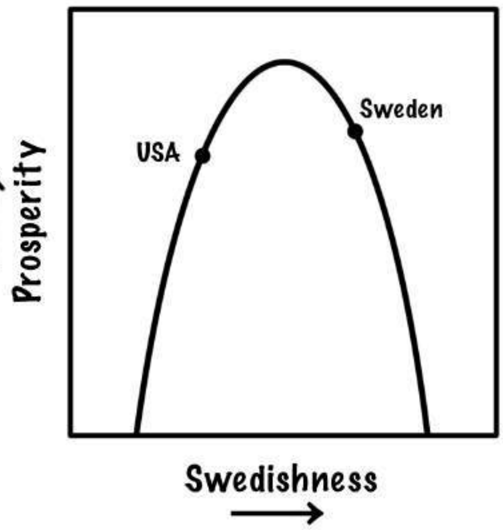
3 Linear Reasoning in Politics

What makes the hottest argument on the Internet? Probably politics. Who completely abuses linear reasoning? Probably political shouters. I do hope that this article receives as much attention as they do, so let's get political. After half of a semester of economics, a good portion of my class now believes that living anywhere beats living in America. Going off of an example from Jordan Ellenberg's book, *How Not to Be Wrong*, I'm sure there's at least one person in my class who is ready to move to Sweden. Ellenberg brings up the battle over the Affordable Care Act, when a title emerged asking, "Why Is Obama Trying to Make American More Like Sweden When Swedes Are Trying to Be Less Like Sweden?" Ellenberg maps the post's line of thought using the diagram below:



At first glance, the title might've gotten you. Yet after a moment of thought, we find many complications, which has to do with our conversation about linear reasoning. Although it seems logical at first, when the scope of the topic gets bigger, we're no longer dealing with lines but with curves.

The graph below suits Ellenberg's taste much better.



Heading towards each other does not have to mean degradation for either one. Instead, both could be heading towards the tip of prosperity! The idea is probably that Sweden taxes their people too much and the USA not enough, so maybe both countries should balance out a bit. I'm not very sure what the situation is with Sweden overall, but hey, we're here for the math anyway. The curve above is also known as the Laffer curve. Named after economist Andrew Laffer, it is the visualization of a theory that deals with how tax rates and the amount of tax collected by the government affect each other. If, like Sweden in this diagram, people get taxed on the higher side, the consequence is that there will be a decreased amount of taxed activities, resulting in less government revenue. The entire theory depends on the assumption that people will change the way they spend or save due to the effects of government services paid by income tax rates.

4 Conclusion

Many lines of writing about lines later, one may or may not be thinking about all the embarrassing times one overused linear reasoning in a battle of words. From whole numbers to Laffer curves, we've gone a long way. In what shapes do you think in?

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Mind Over Matter:
Using Artificial Intelligence to Predict the Shape of Proteins

Ruby Lin

1 Introduction

Stuyvesant requires that of its students who want the Stuyvesant Diploma, i.e. the very small sticker on our regular high school diplomas, to take at least one semester of Computer Science. For those who want to major in biology, the class may seem like a waste of time. But before you run off to your friends and complain about how much you hate the class, consider the possibility that Computer Science can hold the key to future advancement in medicine. Artificial intelligence (AI) was able to predict the structure of proteins' 3D shape better than any other prediction mechanism that exists to this day.

2 A Brief History

Since the 1960s, researchers have been scratching their heads to try and find methods to predict a protein's 3D shape. At first they thought that if they could work out all the individual interactions within a protein sequence, they could predict the 3D shape. As researchers know now, there are hundreds of amino acid sequences per protein and numerous ways each pair of amino acids interact, making the number of possible structures per sequence astronomical. Many researchers tackled this problem, but progress was slow.

3 Current Methods and Their Limitations

Some experimental techniques such as x-ray crystallography and cryo-electron microscopy (cryo-EM) have emerged to help predict the 3D structures of proteins. X-ray crystallography works by obtaining a three-dimensional molecular structure from a crystal. A purified sample of a protein, for instance, is crystalized, and then an x-ray beam would shine through it. The diffraction patterns that result can later be used to analyze the structures that make up the crystal and in turn, the shape of the protein. Cryo-EM is a special microscope that can see the shape of complex proteins at molecular resolution of a frozen hydrated specimen. These methods take up to months or years and don't always work. As of right now, we know the shape of only about 170,000 of the more than 200 million proteins discovered across life forms.

4 Critical Assessment of protein Structure Prediction (CASP) Competition

In 1994, John Moult, and his colleagues started CASP, which takes place every two years, to find potential new ways to predict protein structure. Entrants are given amino acid sequences for about 100 proteins whose structures are not known. For each sequence, certain groups compute a structure, while others determine it experimentally. The organizers then equate the computational predictions to the lab results and assign a global distance test (GDT) score to the predictions. Scores of 90 or higher on a scale of 0 to 100 are considered comparable to experimental methods. When the competition first launched, many groups were able to predict the structures for small, simple proteins. For larger, more challenging proteins, however, GDT scores were as low as 20.

5 DeepMind's AlphaFold

DeepMind first competed in CASP in 2018 with its algorithm called AlphaFold. AlphaFold used a comparative strategy. AlphaFold also incorporated a computational approach called deep learning in which the software is trained on vast data troves to learn patterns. In this case, the vast troves were sequence and structure proteins. However, at that point the predictions were still too coarse to be helpful. To improve, they combined deep learning with an attention algorithm that would mimic the way a person would solve a jigsaw puzzle: first connecting pieces in small clumps then searching for ways to join the clumps into a larger whole. In this case, amino acids were first searched in large clumps then they searched more ways into joining the clumps into a larger whole. Working with a computer network built around 128 learning processors, they were able to train the algorithm to know all 170,000 protein structures. In the CASP contest, DeepMind received a GDT median score of 92.4, higher than any other competitors ever before, paving a way for more innovation.

6 More on how AlphaFold Works

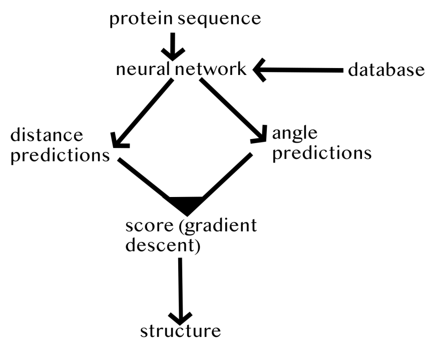
AlphaFold was designed to use neural networks to predict physical properties of a protein from its genetic sequence. The properties which DeepMind's networks predicted were the distance between amino acids and the angles between chemical bonds that connect the amino acids. DeepMind

trained a network to predict the distribution of distances between every pair of residues in a protein. The probabilities were then combined into a score that estimates how accurate a proposed protein structure was. DeepMind also trained another neural network that uses all distances in aggregate to estimate how close the proposed structure is to the right answer.

Using the score functions, DeepMind researchers were able to search the protein landscape to find structures that match their predictions. Since DeepMind built their first method on techniques commonly used in structural biology, they trained a neural network to invent new fragments, which were continually used to improve the score of the proposed protein structure.

The second method optimized scores through a gradient descent. Gradient descent is a mathematical technique commonly used in machine learning for making small, incremental improvements. (Gradient descent is based on the observation that if the multivariable function $F(x)$ is defined and differentiable in the neighborhood of point a, then $F(x)$ decreases fastest if one goes from a in the direction of the negative gradient of F at a, $-\nabla F(a)$.)This technique was later applied to entire protein chains rather than to pieces that must be folded separately before being assembled into a larger structure, to simplify the prediction process.

This entire process can be visualized by this flow chart:



The AlphaFold version used at CASP13 is available here:
https://github.com/deepmind/deepmind-research/tree/master/alphafold_casp13 for anyone interested in learning more.

7 The Application to Medical Advancement

As you may know, proteins are large complex molecules that are essential to everyday life. Nearly every function our body performs relies on proteins and how much they move or change. Each protein's shape determines what its "job" is inside the human body. Since we are in a pandemic, let's use antibodies as an example. Antibody proteins are Y-shaped and form hooks. Because of these hooks, antibodies are able to latch on to viruses and bacteria in order to detect and tag them. Other proteins, like collagen protein, are shaped like cords. Their shape allows them to transmit tension between cartilage, ligaments, and bones which helps hold the body together and enable it to function properly. There are tens of thousands of proteins in the human body alone, each serving a specific purpose, and the recipe to make all those proteins is found in genes.

DNA uses genes in order to code for and make proteins. And just as a missed parentheses while writing code can be catastrophic to the program, an error in the genetic recipe can lead to the malformation of proteins, which can lead to disease or death of an organism.

For instance, cystic fibrosis is a genetic disorder caused by a gene mutation that often removes the amino acid phenylalanine in the 508th position of the cystic fibrosis transmembrane conductance protein. This one deletion of an amino acid causes an individual with this disease to have a faulty channel protein. Without a working channel protein, the body's salt and water concentration gets thrown off, leading to sticky mucus that clogs airways. The expected life span of a person with cystic fibrosis is about 44 years old which is extremely short compared to the norm.

Although scientists know which genes make which proteins, they don't know the shape of the protein and how exactly it folds. If scientists found a way to predict a protein's specific shape, drugs could be created to match a protein's specific shape and help treat diseases like cystic fibrosis.

There is little or no way scientists can predict the shapes of every single protein in an efficient amount of time which is why precision medicine has been so limited. However, programs like DeepMind's AI algorithm, AlphaFold, completely change the game. With the use of previous genomic data and the use of neural networks, AlphaFold was able to predict the shape of complex proteins better than any previous technique, accurately and efficiently. Knowing protein shape can help scientists and researchers create more targeted drugs and therapies and treat disease and fatal conditions caused by a malformation of a protein, saving and improving human lives.

8 Final Thoughts

After researching for hours for theoretical ways to cure genetic disease as a part of my genetics class, I never considered predicting a protein's shape and creating a drug tailored to that shape. I thought cutting out the faulty DNA and replacing it with a new one would be the way to go. Even though I knew that cutting out DNA was risky, that was the only way I could think of to "cure" a disease. I thought that researchers could always just create more precision medicine tailored to a person's individual genetics if cutting and replacing DNA were not an option. Creating drugs to help a protein specific shape is theoretically safer since the person's original DNA isn't tampered with. Without knowing the protein's shape, scientists have spent too much time trying to fix the root of a faulty protein rather than just treating the protein itself. With these new AI programs, computer science can fundamentally change the future of precision medicine and its potential applications.

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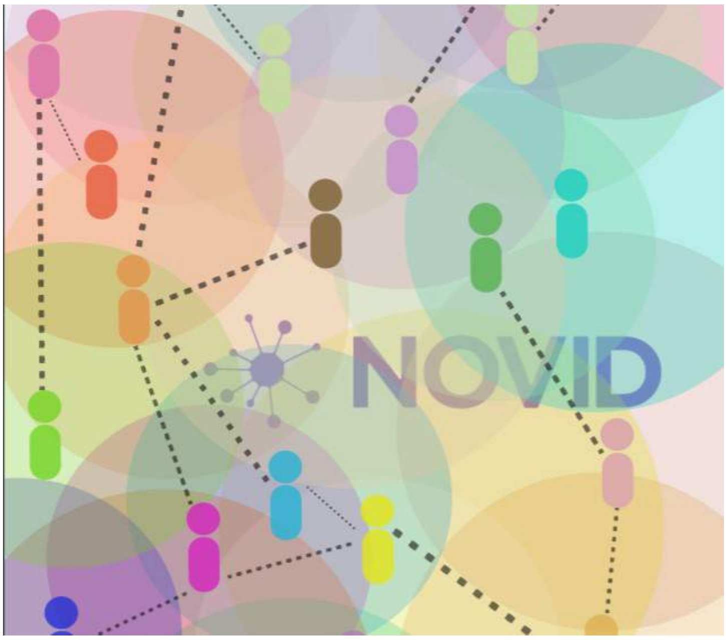
NOVID:
Exploring the Network Theory
Used to Track the Spread of COVID-19

Richard Wang



1 Introduction

NOVID was the first contact tracing app in the world to anonymously trace users' exposure to COVID-19. But how did the Carnegie Mellon University team led by professor Po-Shen Loh develop NOVID using network theory?

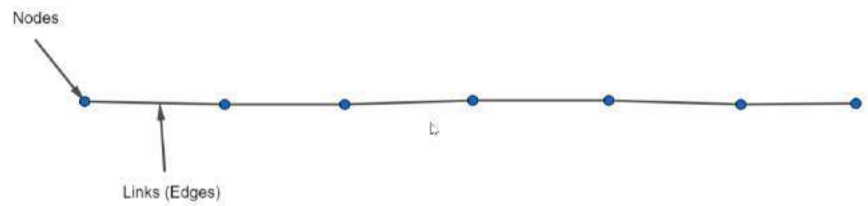
2 Network Theory

Network theory is a technique used to analyze graphs by turning them into complex systems. A network is a graph with two components, the node and the edge. A node (vertex) is a common point connected by branches. In this case, nodes are people. A branch is a line segment that connects two nodes. Edges (links) are lines joining a pair of nodes. Degrees are the number of edges that connect to a vertex. There are multiple kinds of graphs, but only simple graphs are used in this article.

We can analyze the math behind the spread of COVID-19 by considering three different situations.

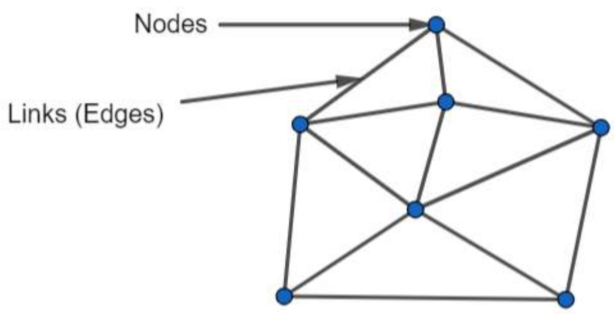
2.1 Situation 1: Weird Society

Imagine a society where the people are organized in a straight line and only contaminate the people next to them– two people in the middle or one person on the edge. The spread would be linear because an infected person can only contact one person on either side at a time.



2.2 Situation 2: Normal Society

However, in a normal society where people spread diseases to others closest to them, growth is exponential. Given a random set of points linked with each other, COVID-19 spreads to the points closest. It takes the second closest point a degree of two to spread COVID-19 to them. It would take the third closest point a degree of three to spread COVID-19 to them, etc. However, the graph also depends on how long people spend with others. In order to stop the spread, links must be closed.



2.3 Situation 3: Accurate Society

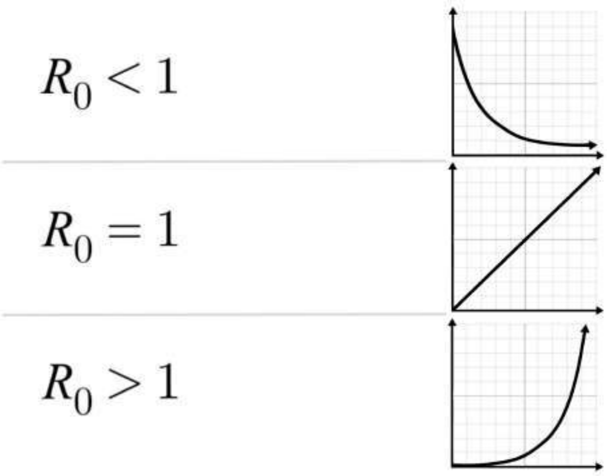
What if the graph were not only based on how close people are to others, but also on how long people are in contact with others? That would be a much more accurate analysis of the network for how COVID-19 spreads. Therefore, links that aren't limited by the nodes they are next to can give even more information.

Let's say that you are notified whether there are any infected points nearby you or if you are infected. The more time people spend with others, the more the COVID-19 infection spreads and the probability of you getting infected increases.

3 Graphing

R0, or R naught, which represents the reproduction number, shows the growth, linear, or decay pattern of a graph. It can be simplified into the following information:

If R0 is less than 1, the graph shows decay. If R0 is equal to 1, the graph is linear. If R0 is greater than 1, the graph is exponential.



In terms of disease transmission, R0 indicates how contagious the disease is. It quantifies the number of people to which an infected person can spread the disease over time.

If R0 equals 1, each existing infection causes one new infection. The disease will stay alive and stable, but there won't be an outbreak or an epidemic. These circumstances were described in Situation 1. If R0 is greater than 1, each existing infection causes more than one new infection. The disease is transmitted between people, and there may be an outbreak or epidemic. This is the case outlined in Situation 2.

The R0 for COVID-19 in April 2020 was 5.7, according to a study published online on *Emerging Infectious Diseases*, an open-access, peer reviewed journal of the Center for Disease Control and Prevention (CDC). That's about double an earlier R0 estimate of 2.2 to 2.7. The 5.7 means that one person with COVID-19 can potentially transmit the coronavirus to 5 to 6 people, rather than the 2 to 3 researchers originally hypothesized. Researchers calculated the new number based on data from the original outbreak in Wuhan, China. With an R0 of 5.7, at least 82 percent of the population needs to be immune to COVID-19 to stop its transmission through vaccination and herd immunity.

The image above shows the actual recorded data of how COVID-19 spread throughout China in early 2020.

How can we use this to help us stop the spread of COVID-19? In terms of network theory, we want R0 to be less than 1, so that each existing infection causes less than one new infection. Therefore, the infection would eventually decline and die out.

4 Contact Tracing

Contact tracing quantifies the following factors to decrease the infection rate:

4.1 Infectious Period

The infectious period is the amount of time a person has been in contact with other people. The longer the infectious period, the greater probability of infection, and the higher the R0.

4.2 Mode of Transmission

The mode of transmission is how the disease is spread. Because COVID-19 is an airborne infection, it spreads more quickly than other diseases that depend on physical contact to be transmitted. Therefore, COVID-19 has higher R0 values than many other types of infections.

4.3 Contact Rate

The contact rate is how many people an infected person has been in contact with over time. This variable is not specific compared to the last two factors because it depends on multiple factors like location, public health measures, quarantines, and travel bans.

In short, contact tracing involves asking a person whether they're infected and if they've been with others who are infected. If either choices are true, quarantine them. From the principle idea of contact tracing, it is helpful for people to be notified whether they are possibly infected. It is to protect other people from you, and to protect you from others.

Contact tracing relies on network theory. If any node is close to another node, the first node can be notified to avoid the others in hopes of safety. This cancels any connections (edges) and isolates every node from risk of infection.

The risk of spreading COVID-19 can be reduced by notifying people to distance themselves from those who are infected and to distance from others if they are infected themselves.

5 Bluetooth-based apps

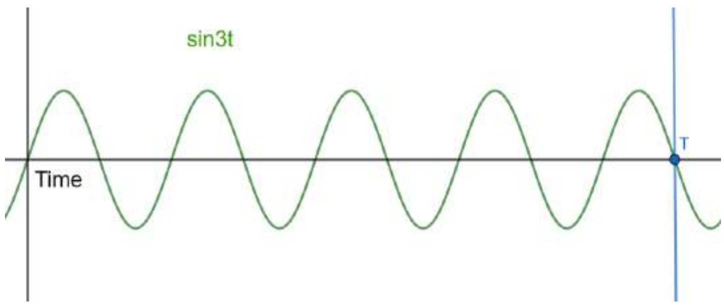
How can we use the information gained through contact tracing and network theory for practical use? We could hire volunteers to record whom infected people have contacted, but it is very inconvenient to record every single infected person. We can allow people to record and survey daily who they've contacted, but then again, it won't be convenient.

The best and most effective way to make contact tracing possible for everyone is by creating a mobile app. The contact tracing app NOVID uses sound waves and Bluetooth to calculate how far users of the app are from other users. This is extremely helpful in fighting off COVID-19, because there is no other contact tracing app in the world that directly tells you how far you are from others.

How does the app work? Given two phones, both phones open their mics. One sends a sound signal to the other and the second phone closes their mic and analyzes how long it took to receive the sound. The app notifies whether or not someone is at a maximum of 9 feet away. The 9 feet takes into account whether or not you're holding your phone, which decreases the distance between two phones due to arm length.

5.1 Mathematics of the Sound Signal

Although the app can potentially prevent COVID-19 infections, there are still imperfections to the app. If, for example, you are in your house, the sound can bounce off the walls and need the structure of the room in order to calculate how the sound signals are traveling. In addition, the phone must pick the proper frequency of the sound signal. Unfortunately, the mathematics in how sound signals are calculated despite being bounced off of surfaces is not publicly displayed. However, the mechanics for finding the proper frequency can still be calculated.



The picture above gives an example of estimates for what a phone would pick up if it is expecting a signal. How can we tell that this signal is the same signal from the NOVID app? In other words, how can we tell that this is the proper frequency the phone catches?

In the graph, the function of the graph is $sin3T$. If we calculate the integral (area under the curve) of the graph from 0 to T, then the area and T have a linear relationship. What if we calculate the integral of $f(x)sin2T$ from 0 to T? You can calculate it by substituting as follows:

$\sin 3t * \sin 2t = 1/2[\cos(3t - 2t) - \cos(3t + 2t)] = 1/2[-\cos 5t]$ which becomes

$1/2 * \int_0^T \cos(t)\, dt - 1/2 \int_0^T \cos(5t)\, dx$.

This value approaches a constant. Therefore, we can derive any sound signal with the same frequency by calculating whether or not the function of the signal has the same value as the constant.

6 Conclusion

In conclusion, with the help of mathematics, the NOVID app was developed as an effort to limit COVID-19 infections. From network theory to analyzing different social models to graphing the spread of COVID-19 to finding the factors of infection then to how to stop the spread by using an app, a web of relations emerged from the mathematics we study at school.

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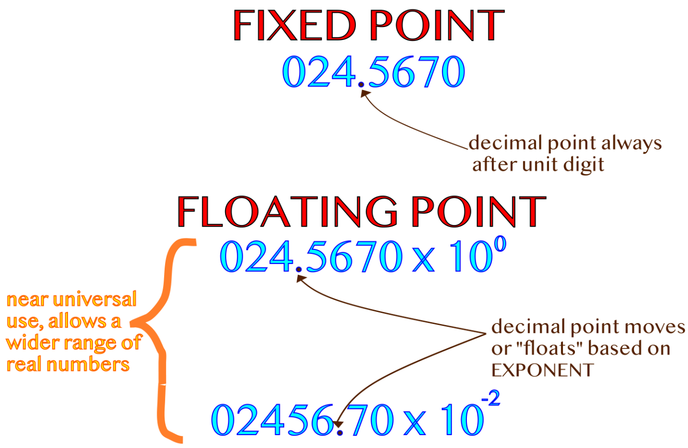
Getting Right to the Point: Approximations of Floating Point Numbers

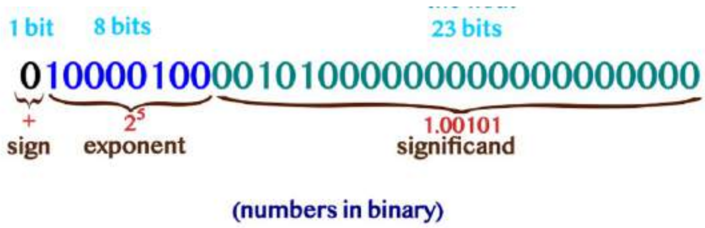
Jeremy Ku-Benjet

Long ago, 16-bit microprocessors were state of the art. Each computer came with its own way of representing real numbers. Floating point numbers, numbers where the decimal point can move, were almost universally preferred.

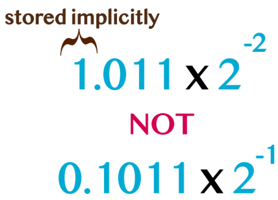


However, different implementations of floating point numbers or “floats” each came with their own quirks. Suffice to say, when subtracting two distinct numbers results in zero in some, but not all, implementations of floating point arithmetic, there is a problem. To combat the chaos, there needed to be standardization, and so efforts to unify floats under the name of the IEEE (Institute of Electrical and Electronics Engineers) began in the late 1970s. Meetings included representatives from Motorola, Intel, National Semiconductors, and many of the other major microprocessor manufacturers. It was a unified effort to agree on a good standard for floating point numbers. Of course, there was still strong debate over the details of the standard, delaying its creation for several years. Slowly, the fundamentals of the new standard were agreed upon, and by 1984 it was implemented nearly universally by microprocessor manufacturers. The following year it was officially published as IEEE 754 and since has been used, not just for hardware, but for representing numbers in software as well. In the most common representation, a floating point number is structured the same

as a number in scientific notation. There is a sign, some significant digits, called the significand (or mantissa), and an exponent. Computers find it convenient to use base 2, so it will be used for the remainder of this article.



According to IEEE 754, a single precision floating point number is represented by 32 bits. The first bit is the sign bit, 1 if the number is negative, else 0. Following is the exponent, an 8-bit positive integer, 127 more than the exponent of the 2. Storing the exponent with this "bias" allows for negative exponents and therefore numbers between 1 and -1. The final 23 bits are the significand (or mantissa). They describe the details of a number. So there are no leading zeros, which could more efficiently be stored in the exponent, the significand is "normalized." If the significant digits of the number are 1.00101 then the significand is 00101. The 1 in the significant digits is implicitly stored. As the number is normalized, the 1 is always present so storing it in the significand is needless.



There are some special floats which don't follow the normal format, such as zero, but listing them here would be superfluous.

Integers can be converted into floats. A program can easily figure out the sign, exponent, and significand; however, almost any modern computer will have an instruction to convert an integer into a float faster than could be written otherwise. To find the exponent of the float, the computer computes $2n$, which can be useful to know. Therefore, by taking the second through ninth bits of a float, converting an integer to a float can be used to quickly get the binary logarithm of a number. The limitation of this approximation is precision. The method to get more precision is to take n to a power:

$$\frac{\log_2 n^m}{m} = \frac{mlog_2 n}{m} = \log_2 n$$

A larger input to the $\log_2$ means the integer portion of the output will be larger and the final output can be made more precise:

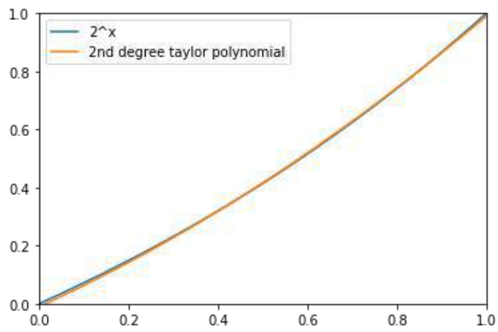
$$[\log_2 10] = 3 \text{ and } \frac{[\log_2 10^1 0]}{100} = 3.3$$

However, the powers of the input grow exponentially and can quickly become too large to be easily handled. Getting another decimal digit of precision requires handling 10^{100}, a number too large to be stored in a standard 4-byte or 8-byte integer.

A float does get around having to take the binary logarithm; it just passes the operation off to a faster part of the program. Converting an integer to a float and reading the exponent to approximate a logarithm suggests writing to the exponent and interpreting the data as an integer, in a way, the inverse of how a logarithm was approximated. This method could also be used to approximate an exponential. This becomes clear, as the structure of 2^x, when broken into a fractional and integral part, mirrors a float's representation. Breaking up x into $[x]$ and x, the integer part and the fractional part of x respectively, $2^{[x]+x} = 2^{[x]} \cdot 2^x$. $2^{[x]}$ is an integer, with $[x]127$ less than the exponent of the float (due to the bias), and $0 \leq 2^{[x]} < 2$ is a fraction, resembling the significand.

$$2^x = (1+\underbrace{2^{\{x\}}-1}_{\text{approximated}}) \times \underbrace{2^{(\lfloor x \rfloor+127)-127}}_{\text{computed when reinterpreting the float}}$$

The only unknown information about the float representing 2^x is the significand, $2^x - 1$. The minus 1 is due to the implicit 1 stored in a float. Therefore, the significand must be approximated. However, approximating $2^x - 1$ is much easier than the original problem of approximating 2^x as $0 \leq x < 1$. The range {x} is relatively small, so using a Taylor polynomial to accurately approximate all $2^x - 1$ is possible.



Combining the calculated exponent and approximated significand into a single float creates an efficient method to take 2 to any power.

Writing $n = (1 + s)2^e$ where n is a number, s is the significand, and e is the exponent, and $N = (1+s)2^{23}+e*2^{23}$, where N is interpreting the float representation of n as an integer, allows for alternate approximations of exponents and logarithms. Note the 2^{23} in the integer interpretation of the float comes from the significand being 23 bits long.

Taking $\log(n) = \log(1 + s)2^e$ and using N to solve for e allows for $\log(n)$ to be rewritten in a more convenient form such that only $\log(1 + s) - s$ must be approximated. As s can only range from zero to one, this approximation is possible with accuracy.

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Solving Fermat's Last Theorem: Andrew Wiles' Breakthrough After Over 350 Years

Lamisa Aziz

1 Pierre de Fermat

Pierre de Fermat was a famous 17th-century French mathematician whose nickname is "the founder of the modern theory of numbers." He is often known for his early developments that led to modern calculus and his contribution to the early progress of probability theory. Being inspired by the "Arithmetica" of the mathematician Diophantus, he continued to discover various new patterns in numbers which led to the creation of many new conjectures and theorems, but the most unique one was his last theorem.

2 Fermat's Last Theorem

Fermat's Last Theorem was called his *pièce de résistance*, a conjecture he was working on but left unproven until the day of his death. Fermat's mathematical work was usually communicated through letters to his colleagues or friends but with little or no proof. Even though he claimed to have proved all the arithmetic theorems, there are only a few records left to actually show it. His unfinished last theorem puzzled mathematicians for over 350 years. The theorem was originally described in a scribbled note in the margin of his copy of Diophantus' "Arithmetica." Because there was no proof for his theorems, many mathematicians began to doubt his claims, especially since the conjecture has been found to be one of the hardest mathematical problems in the world to prove.

Fermat's Last Theorem makes the statement that there are no natural numbers $(1, 2, 3, ...)$ $a, b,$ and c such that $a^n + b^n = c^n$, in which n is a natural number greater than 2. For example, if $n = 3$, Fermat's Last Theorem states that no natural numbers $x, y,$ and z exist such that $x^3 + y^3 = z^3$.

3 Proof for Fermat's Last Theorem

Countless mathematicians were confused about how to finish his proof. There were proofs for many different parts of the theorem; many for specific values of n were devised. For example, Fermat himself did a proof of another theorem that effectively solved the case for $n = 4$, and by 1993, with the help of computers, it was confirmed for all prime numbers $n < 4,000,000$. By that time, mathematicians had discovered that proving a special case of a result from algebraic geometry and number theory known as the Shimura-Taniyama-Weil conjecture would be equivalent to proving Fermat's Last Theorem.

However, one mathematician in particular named Andrew Wiles broke new ground. He was an English mathematician who had been interested in the theorem since he was 10 years old and finally presented the proof of the Shimura-Taniyama-Weil conjecture in 1993. When revised, there was an error that was found in this proof, but with help from his former student Richard Taylor, Wiles finally devised a proof of Fermat's Last Theorem. His work of the proof was published in 1995 in the journal *Annals of Mathematics*, and the Norwegian Academy of Science and Letters awarded him with the 2016 Abel Prize "for his stunning proof of Fermat's Last Theorem by way of the modularity conjecture for semistable elliptic curves, opening a new era in number theory."

4 Wiles' Proof

Wiles' proof has three main parts to it. The first part is setting up for the proof. We start by assuming (for the sake of contradiction) that Fermat's Last Theorem is incorrect. That would mean there is at least one non-zero solution (a, b, c, n) (with all numbers rational and $n > 2$ and prime) to $a^n + b^n = c^n$. Then we use Ribet's theorem (using Frey and Serre's work) to show that we can create a semi-stable elliptic curve E using the numbers ($a, b, c,$ and n), which is never modular. If we can prove that all such elliptic curves will be modular (meaning that they match a modular form), then we have our contradiction and have proved our assumption (that such a set of numbers exists) was wrong. If the assumption is wrong, that means no such numbers exist, which proves Fermat's Last Theorem is correct.

The second part is the Modularity Lifting theorem where Wiles aims to first prove a result about these Galois representations of elliptic curves $\rho(E, p)$ for any prime $p > 3$, that he will use later: that if a semi-stable elliptic curve E has a Galois representation $\rho(E, p)$ that is modular, the elliptic curve itself must be modular. By proving this, it makes counting and matching easier and proves the representation is modular. Then Wiles used proof by induction and a class number formula ("CNF") to show that once the hypothesis for one elliptic curve is proved, it can automatically be extended to be proven for all subsequent elliptic curves. This shows a key point about Galois representations which is that if the geometric Galois representation $\rho(E, p)$ of a semi-stable elliptic curve E is irreducible and modular (for some prime number $p > 2$), then subject to some technical conditions, E is modular. This is Wiles' lifting theorem which he continues to part 3 where he finally proves that all semistable elliptic curves are modular using the modularity lifting theorem. The final result proved that whether or not $\rho(E, 3)$ is irreducible, E (which could be any semi-stable elliptic curve) will always be modular. This means that all semistable elliptic curves must be modular.

June 23 marks the date where Andrew Wiles proved Fermat's Last Theorem, the most famous math problem at the time. After Wiles had completed this great achievement, there were more ideas that talked about a new golden age of mathematics in number theory, which the Last Theorem was known for. While his proof was published, there were still many who questioned him and his answers. One such misunderstandings was in Fermat's Last Theorem which said, "If n is any positive integer greater than 2, then it is impossible to find three positive numbers a, b and c such that $a^n + b^n = c^n$. This idea contrasted with what happens to the Pythagorean triples when $n = 2$ like $3^2 + 4^2 = 5^2$, where the rule does work and isn't impossible. Mathematicians tried to explain this contrast to the best of their ability, but it just led to more questions. Eventually, a computer scientist provided a logical analysis of Wiles' proof, and this automated proof of verification helped give the mathematicians some solutions to their questions. The verifications involve reformulating the proof as a series of statements, each written in a language understood by the computer, which then confirms every step and stamps the constitutional certification. This method is usually applied to long and extremely difficult proofs.

This method helped them figure out that if there is a contrary to Fermat's Last Theorem where there is a triple of positive integers $a, b,$ and c that $a^p + b^p = c^p$, for an odd number p where this always stood true. Otherwise, $a, b,$ and c could be rearranged into a new equation called an elliptic curve. The elliptic curve is shown into a Galois representation which consists of an infinite collection of equations that list the properties that were expected to be impossible. Every Galois representation should be assigned by a modular form. Yutaka Taniyama and Goro Shimura are credited with the proof for this solution. Wiles had only proved a small part of this modularity conjecture, and it was finally established by other mathematicians like Christophe Breuil, Brian Conrad, etc.

5 Conclusion

Fermat's Last Theorem is definitely one of the most challenging math problems in history and was only solved because of the hard work of countless mathematicians starting with Fermat, then Andrew Wiles, and leading to mathematicians who try to understand it even today. One application of Fermat's Last Theorem is found in this question: How many ordered triplets exist when you define a primitive inverse-cubed triplet (x, y, z) where the gcd $(x, y, z) = 1$ and $\frac{1}{x^3} + \frac{1}{y^3} = \frac{1}{z^3}$? We know the answer is that no such (x, y, z) exists because by Fermat's Last Theorem, the only solution occurs when $x^2y^2z^2 = 0$. More applications of this important theorem will no doubt be discovered for years to come.

6 Problems for Further Exploration

Problem 1: What is the number of distinct positive integers n such that $n + 3^2$ and $n^2 + 3^3$ are both perfect cubes?

Problem 2: Let n be a positive integer which makes $(4n^2 + 6n + 3)\frac{n}{3}$ a triangular number. How many solutions are there for n?

Problem 3: Find the number of positive integers n such that there exists an integer $m \neq \frac{n}{2}$such that $\frac{n^3-2m^3}{6m}$ is a perfect square.

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Exponential Mayhem

Aditya Pahuja

A mathematician created a calculator with buttons for 10^x, e^x, $\log x$, $\ln x$, and $\lceil x \rceil$ ($\lceil x \rceil$ denotes the least integer that is greater than or equal to x). One day, she was wondering how to compute different numbers using her calculator. How would you express the numbers from 2 to 16 using the available functions?

The Rules: The calculator's screen initially displays 0. Upon being pressed, each button (function) takes the value currently on the screen as its input.

The Puzzle: Assuming that the calculator can calculate logarithmic and exponential functions with infinite precision, your task is to create expressions that equate to every integer between 2 and 16, inclusive. For example, $3 = \lceil e^{e^0} \rceil$.

(Some) Solutions to *Exponential Mayhem*

Aditya Pahuja

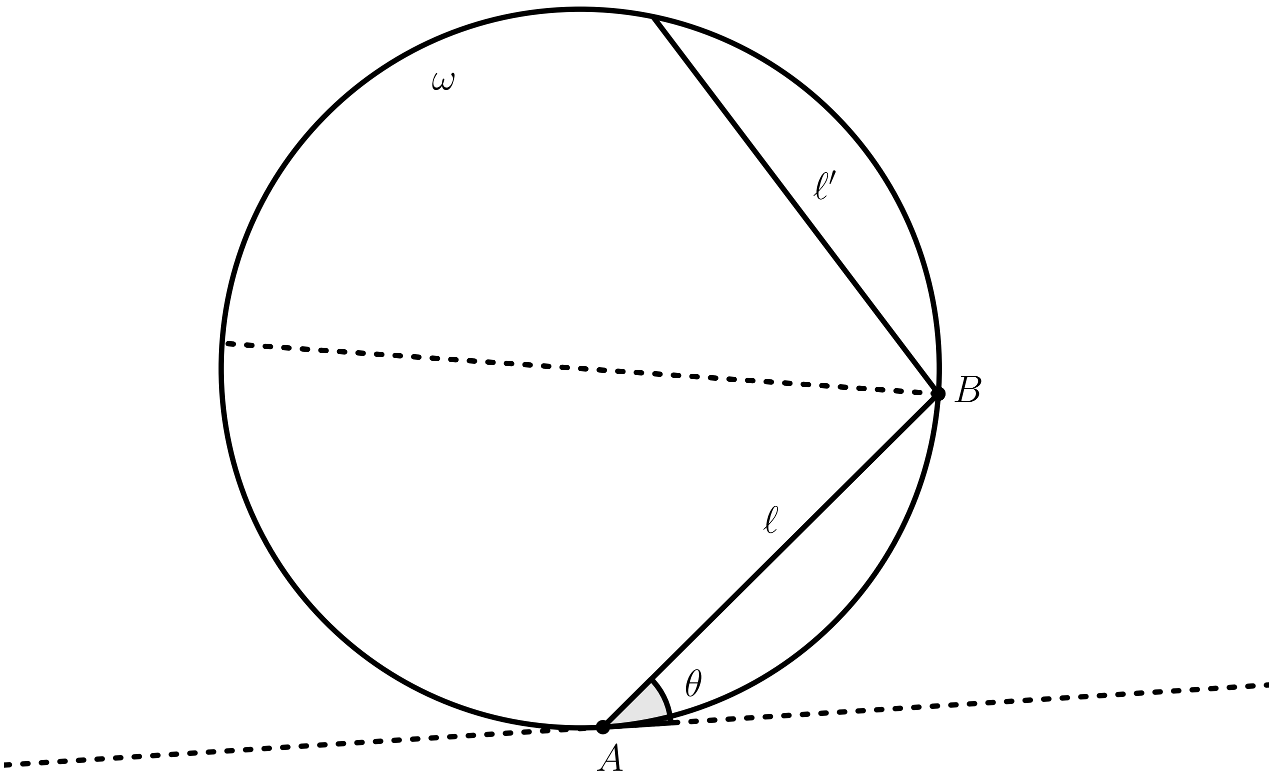
$$
\begin{align}
1 &= e^0 \tag{1}\\
2 &= \lceil \log(e^{e^1}) \rceil \notag\\
3 &= \lceil e^1 \rceil \notag\\
4 &= \lceil e^{\log(e^{e^1})} \rceil \notag\\
5 &= \lceil \log(e^{10^1}) \rceil \notag\\
6 &= \lceil e^{\log(e^4)} \rceil \tag{2}\\
7 &= \lceil \log(e^{e^{e^1}}) \rceil \notag\\
8 &= \lceil e^2 \rceil \tag{3}\\
9 &= \lceil e^{\log(e^5)} \rceil \tag{4}\\
10 &= 10^1 \notag\\
11 &= \lceil \ln(10^{\ln(10^2)}) \rceil \tag{5}\\
12 &= \lceil \ln(10^5) \rceil \tag{6}\\
13 &= \lceil 10^{\ln(3)} \rceil \tag{7}\\
14 &= \lceil e^{\log(e^6)} \rceil \tag{8}\\
15 &= \lceil \log(\log(e^{10^{e^{e^1}}})) \rceil \notag\\
16 &= \lceil e^{e^1} \rceil \notag
\end{align}
$$

Equation 1 can be substituted into any of the equations that have a 1 in the exponent; it is just neater to not have to write e^{e^0} for e. Additionally, equations 2 to 8 use numbers that are not equivalent to 0 or 1; it is quite clear that you can simply substitute the expression that equates to that number. For example, in equation 3, you can substitute $\lceil \log(e^{e^1}) \rceil$ for 2. This substitution will result in an expression that follows the constraints of the puzzle.

Reflective Circles

Aditya Pahuja

A laser starts at point A on the circumference of a circle, ω. The laser moves along a chord from point A to some other point B on ω, labeled ℓ. The angle between the line tangent to the circle at point A and ℓ is θ. When the laser reaches point B, it bounces off the circle. It then moves along ℓ', which is the reflection of ℓ over the diameter of ω that passes through point B.

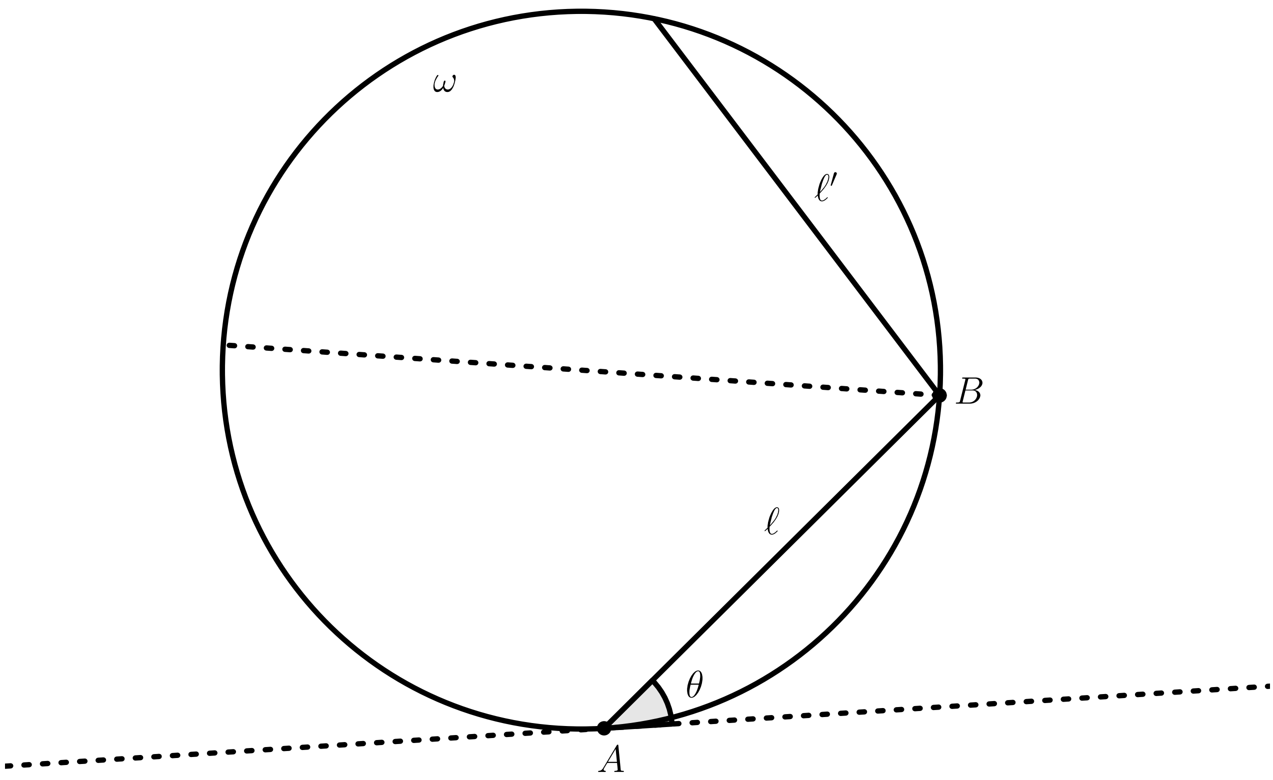


Problem 1: Assuming this process continues indefinitely, for which values of θ does the laser's path reach point A?

Problem 2: What can we say about the set of points on ω that the laser will never reach?

Solutions to Reflective Circles

Aditya Pahuja



Solution 1:
For which values of θ does the laser's path reach point A?

Since the angle made by ℓ and the tangent to point A is θ degrees, arc AB must have a measure of 2θ. ℓ' is a reflection of ℓ, so it is congruent to ℓ. Thus, the arc that ℓ' subtends is congruent to arc AB, which means it has a measure of 2θ as well. Every single bounce made by the laser follows a path that is the result of a sequence of reflections of ℓ, so every single path goes a multiple of 2θ degrees around the circle. Since a circle's circumference covers 360 degrees, the path only ever circles back to point A if there exists an integer n such that $2n\theta$ is a multiple of 360.

If θ is rational, then it can be written as $\frac{p}{q}$, where p and q are relatively prime positive integers. If we let $n = 360q$, then $2n\theta$ is a multiple of 360. If θ is irrational, then $2n\theta$ is also irrational because the product of a rational and an irrational number is irrational, which means that n only exists for all rational values of θ, so the laser circles back to point A if and only if θ is rational.

Solution 2:
What can we say about the set of points on ω that the laser will never reach?

Since the laser travels 2θ degrees along the circle between bounces, every point it hits must be a multiple of 2θ degrees around the circle from point A. Thus, the set of points that it does not reach is simply the set of points on the circle that are not a multiple of 2θ degrees from point A.

To be more specific, we can look at two cases: either θ is irrational or it is rational.

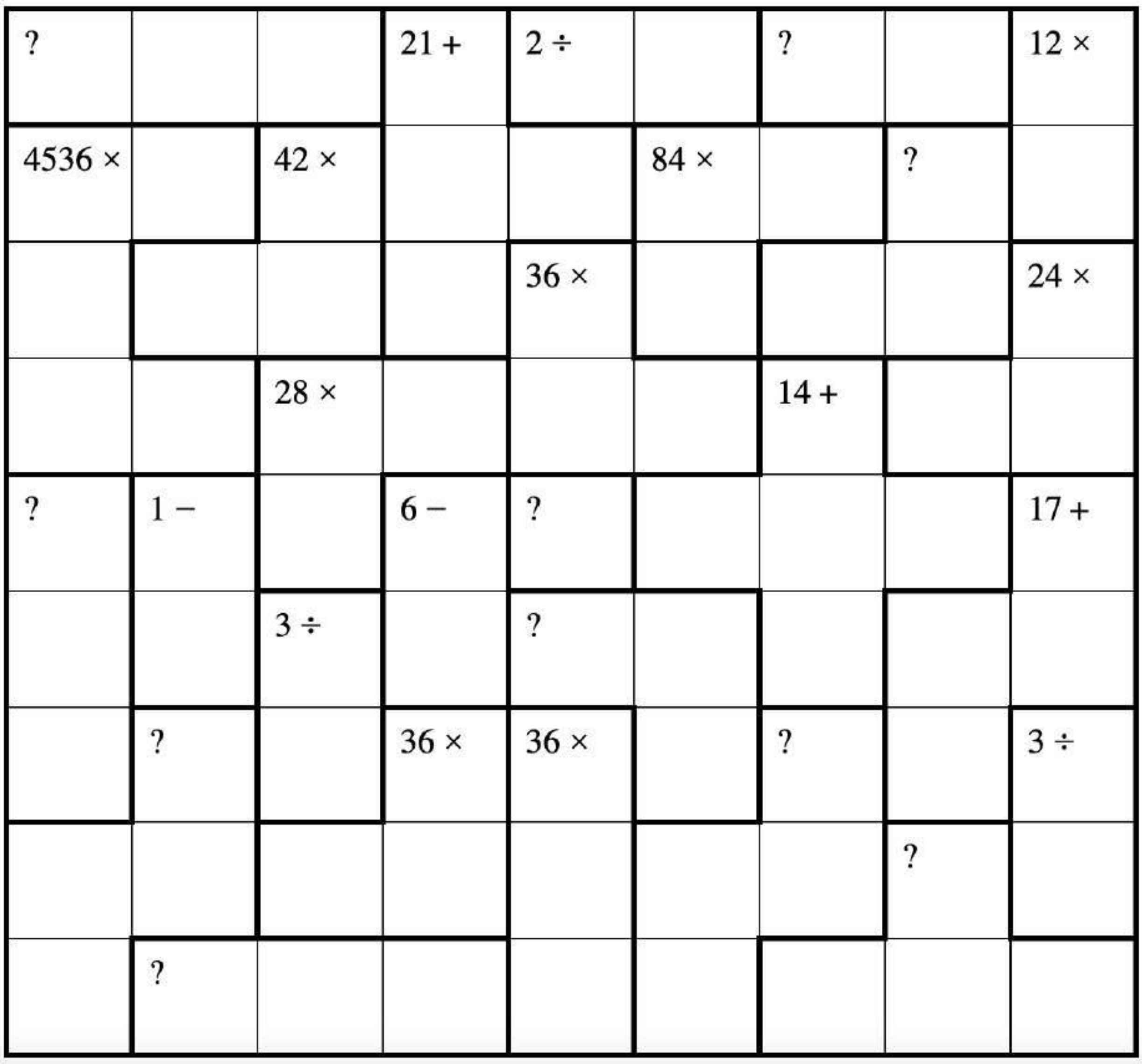
If θ is irrational, then the laser cannot reach any point that is a rational number of degrees from point A (except for point A itself). We can prove this using contradiction. Assume that there is a point that is $\frac{p}{q}$ degrees from point A that is on the laser's path. Then, $\frac{p}{q} + 360k = 2n\theta$, where k and n are nonnegative integers. However, the left side of the equation is rational, while the right side is irrational, which is a contradiction. Thus, the set of points that the laser cannot reach if θ is irrational includes the set of points a rational number of degrees from point A.

Similarly, we see that if θ is rational, then the laser can never reach a point which is an irrational number of degrees away from point A.

"Relish" the Challenge

Edward Wu

Ding! Your hotdog (or veggie dog) has just finished heating in the microwave and you go take it out. Now comes the fun part of slathering your favorite toppings on top while you hold it in one hand: a drizzle of ketchup, a squeeze of mustard, a scoop of onions, a spoonful of relish... Oops! Your relish goes off the side of the bun and onto the paper below. Great, there goes that KenKen I was planning on doing, you think. Since you have been a dedicated environmentalist for a year and going by now, you really don't want that streak to end by printing another because that's a waste of paper and ink. Oh well, I'll give it a shot anyway, you say to yourself. See if you can finish it! (P.S. You can.)



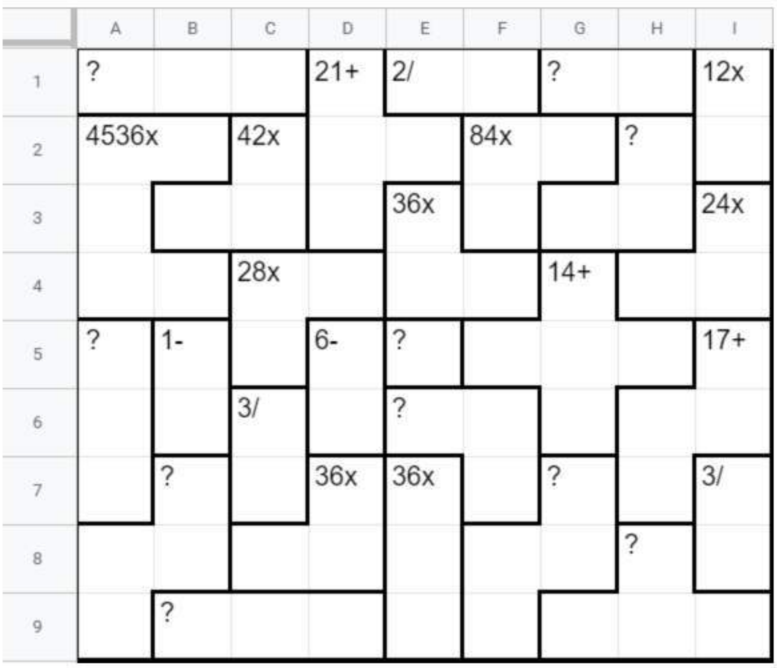
Solution to "Relish" the Challenge

Edward Wu

Preface

Provided is just one of the solutions that you can use. The solution has been broken up into parts based on significant breakthroughs in the puzzle. At the top of each page after this one will be the significant breakthrough(s), followed by the explanation. Sometimes other numbers that can be discovered using this information will also be revealed. There will also be an image of the current board after all these other numbers are inserted into the grid. This way, you can look at the top of each page to get only the next hint.

In the solution, we will define the grid to be the entire 9 x 9 field. Each 1 x 1 subgrid will be called a cell. A cell is denoted by a column, given a letter from A to I, and a row, given a number from 1 to 9. For example, E5 will denote the center cell of the grid. A cell is *solved* when the number that goes inside that cell is limited to one value. A clue represents an arithmetic rule for one or more cells that describes the relationship of the numbers in these cells. A group is a group of cells that share the same clue. Groups are bounded by bold lines in the grid. A cell is said to be *occupied* by a digit when that digit appears in that cell in the final solution. Standard KenKen rules apply.



Step 1: Cell C5 is a 7

Explanation:

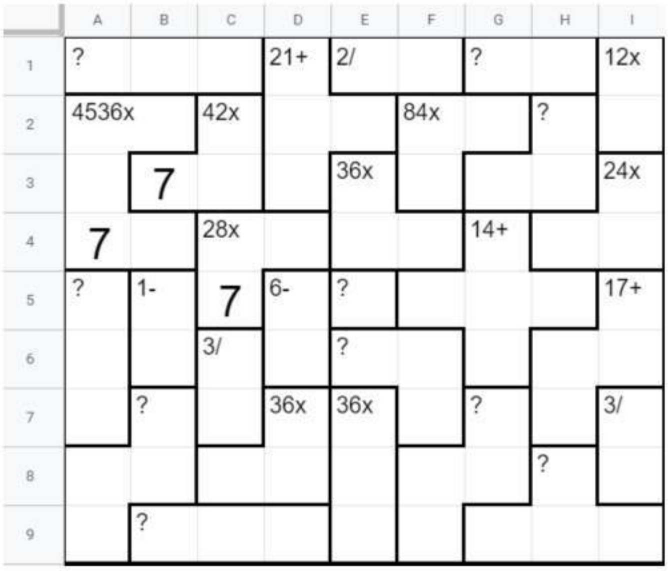
In Rows 2 through 4 are many clues that involve multiplication to a value that is a multiple of 7. Because 7 is prime, a 7 must occupy at least one cell in each of these groups. In Rows 2 and

3 specifically, there is a 42x clue and a 84x clue. These two groups will each require a 7. Because each row can only have one 7, the 7s in Rows 2 and 3 will be used in these two groups.

The prime factorization of 4536 is $2^3 3^4 7$. Since 7 divides 4536, 7 must occupy one of the cells in this group. We just claimed that the 7s in Rows 2 and 3 are used elsewhere, so the 7 in the 4536x group cannot appear in Row 2 or 3. Thus, it must appear in Row 4.

We also see that 7 divides 28. Since the 7 in Row 4 occupies either cell A4 or cell B4, the 7 required in the 28x group cannot appear in Row 4. Then the 7 must occupy the last cell in this group: cell C5.

Follow-up:
Returning to the 42x group, with a 7 occupying cell C5, the 7 in the 42x group cannot be in Column C. Then 7 must occupy cell B3. Likewise, the 7 from the 4536x clue that was originally restricted to Row 4 can no longer occupy cell B4. Therefore, it must occupy cell A4.



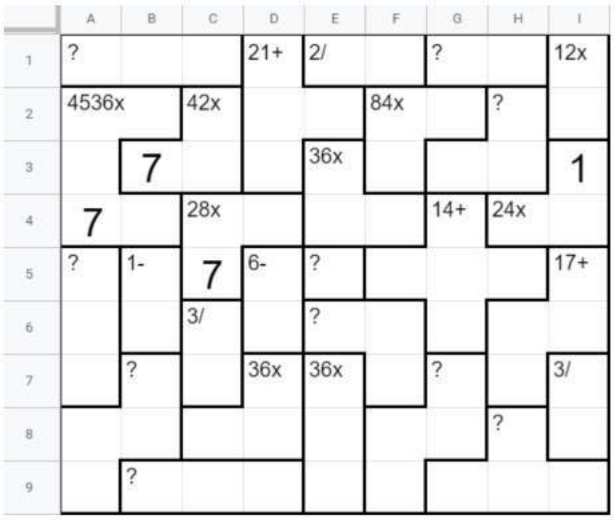
Step 2: Cell I3 is a 1

Explanation:
In Row I there are many clues that can be solved with the same numbers. In particular, these are the 12x, 24x, and 3/ clues. 12x can be solved by (2,6) or (3,4). The 3/ can be solved by (1,3), (2,6), or (3,9). Notice that by the rule that each number must appear exactly once in each column, then the 2, 3, and 6 are all used within these two groups. Imagine that the 12x was solved by (2,6). Then the 3/ cannot be solved by the (2,6). Both (1,3) and (3,9) use a 3, so in the end, 2, 3, and 6 were all used. Similarly, if the 12x was solved by (3,4), then the 3 cannot appear in the 3/ clue, so 3/ must be solved by (2,6). With this in mind, we can force the options for the 24x clue.

Because the 24x clue spans two columns and two rows, we can allow one digit to be repeated once. This gives four options to fill the 24x group: (1,3,8), (1,4,6), (2,2,6), and (2,3,4). Notice that since we already used the 2, 3, and 6 in Column I, we can only have one copy of 2, 3, or 6 in the 24x group because only one cell exists outside of Column I. Then we can eliminate the last two.

Out of the remaining two options (1,3,8) and (1,4,6), a 1 will be present in the 24x clue. Let's return to the 28x clue. 28x can be solved in three cells with (1,4,7) and (2,2,7). Since the 7 solves cell C5, the remaining two digits cannot be the same, because they appear in the same row.

Therefore, 28x must be solved by (1,4,7). Since a 1 is confirmed to appear in either cell C4 or cell D4, then there cannot be any more 1s in Row 4. Therefore, the 1 in the 24x clue must appear in cell I3.



Step 3: Cell E6 is a 7

Explanation:

There are only so many places any particularly digit can be in any row or column. In Column E, 7 cannot be in the 2/ group in Row 1 or the 36x group in Rows 7 through 9. From the 7s we have already deduced to solve cells A4, B3, and C5, there cannot be any more 7s in this row. Therefore, the 7 in Column E cannot appear in cells E3, E4, or E5. In the 84x clue that spans Rows 2 and 3, we need a 7 in this group. There cannot be a 7 in Row 3, so it must appear in Row 2. Then there cannot be any more 7s in this row. In particular, cell E2 is not a 7. This leaves one spot for the 7 in Column E: cell E6.

Follow-up:

We can use similar reasoning with the 7 in Column I to deduce that cell I9 is also solved by a 7. Then cell D1 is a 7. We have thus placed six 7s.

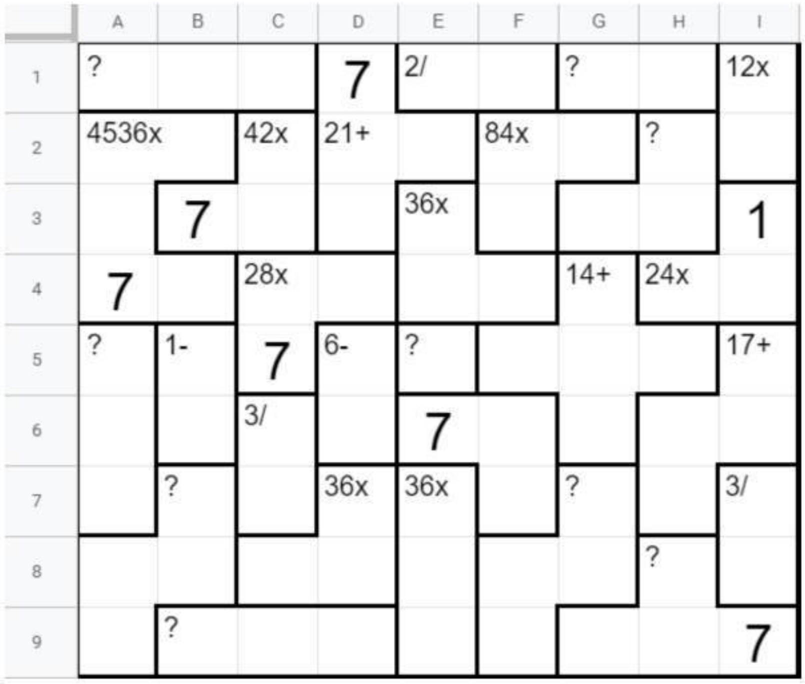
Step 4: Cell B4 is a 9

Explanation:

Consider the 36x group in Column E. All three digits must be distinct because the three cells appear in the same column. There are two ways to solve this clue: (1,4,9) and (2,3,6). Now consider the 36x group in Columns E and F. With the repeat, there are five options to solve this clue: (1,4,9), (1,6,6), (2,2,9), (2,3,6), and (3,3,4).

Let's eliminate some of these possibilities. The 28x group must be solved by a (1,4,7) and not a (2,2,7) because the two empty cells are in the same row. Then there cannot be any other 1s in Row 4. The 1 in Row 3 solves cell I3. Then there cannot be a 1 in the 36x group in the middle. It must be solved by (2,2,9), (2,3,6), or (3,3,4).

Returning to the vertical 36x group in Column E, assume it is solved by (2,3,6). If this were the case, none of the three remaining options for the 36x above it could solve it because there is only one spot for a 2, 3, or 6. For example, if the vertical 36x group was (2,3,6) then the 36x group



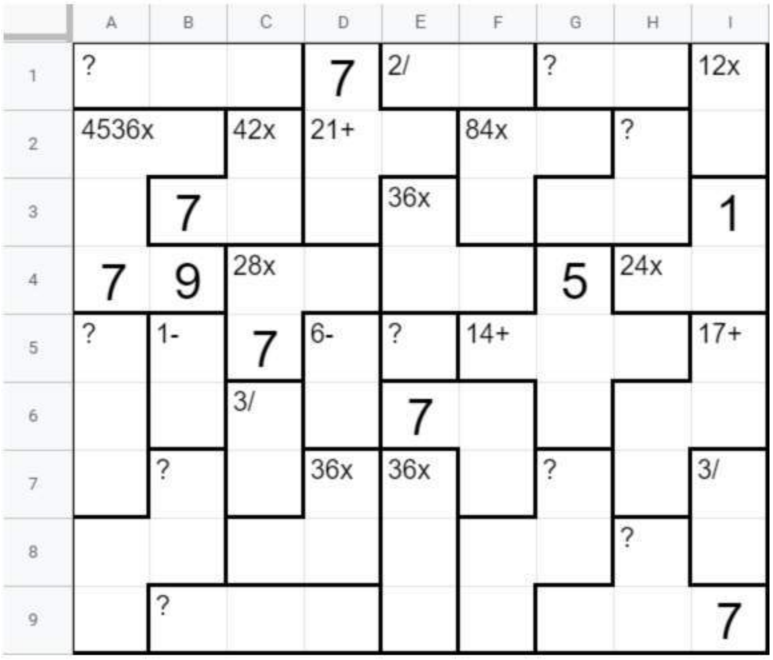
above it cannot be solved by (2,2,9) because the 2 in Column E will be placed in the vertical group, and the two 2s in the middle 36x group cannot both be in cell F4.

Since none of the remaining possible solutions for the middle 36x group will work with the bottom being solved by (2,3,6), the vertical 36x clue must be solved by (1,4,9). With this in mind, the middle 36x group cannot be solved by (1,4,9). It also cannot be solved by (2,2,9), (3,3,4), or (1,6,6) because this would force the 9 or 1 to be in cell E4. Then the middle 36x group must be solved by (2,3,6).

In Row 4, the 9 cannot be in the 28x or 24x clues because of divisibility. The middle 36x clue is solved by (2,3,6) so the 9 cannot solve cells E4 or F4 either. If cell G4 was solved by a 9, then the remaining four cells of the 14+ group must add to 5, which is impossible given its shape. Therefore, the 9 must solve the last remaining cell in Row 4: cell B4.

Follow-up:

Divisibility also shows that cell G4 must be solved by a 5.



Step 5: Cell I4 is an 8

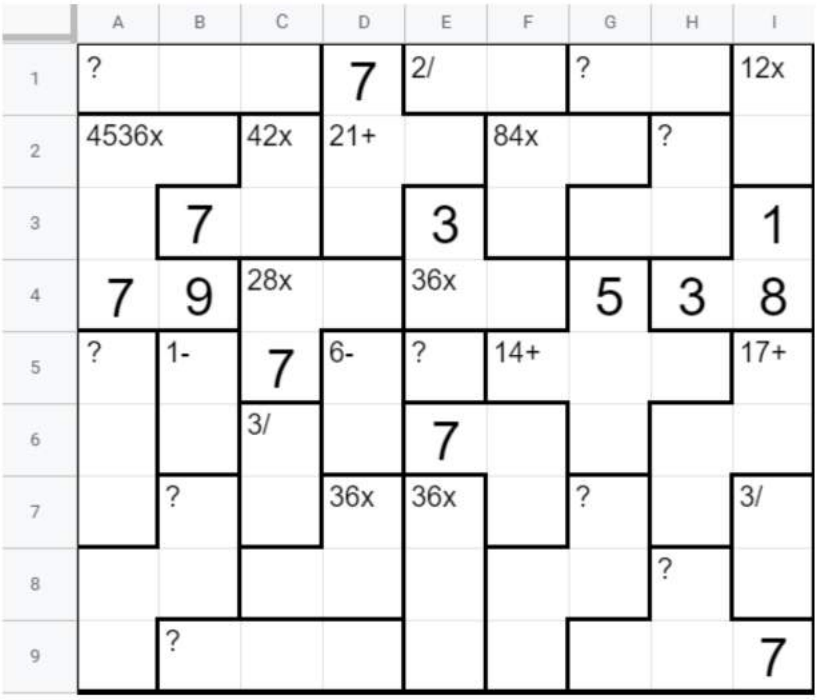
Explanation:

In Row 4, we have solved cells A4, B4, and G4. We know that cells C4 and D4 are (1,4) and cells E4 and F4 are elements of (2,3,6). Since 8 cannot be in either of these groups, it must be in the 24x at right. (This can also be shown using divisibility rules.) If there is an 8 in the 24x clue, then it must be solved by (1,3,8).

In Step 2, we proved that the 2, 3, and 6 in Column I must solve cells in the 12x group or the 3/ group. Then cell I4 must be solved by the 8 and cell H4 by the 3.

Follow-up:

We see that 3 solves cell H4. Then cells E4 and F4 are (2,6), leaving another 3 for cell E3.



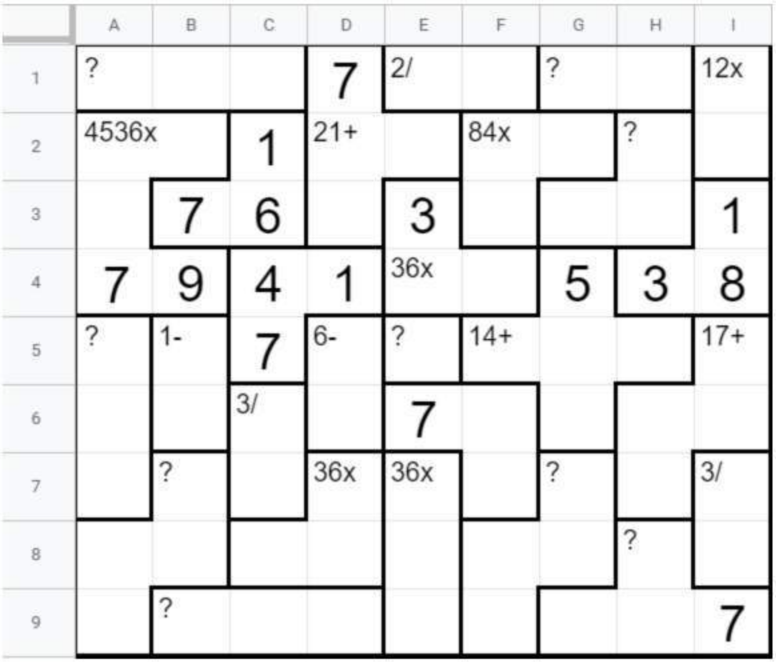
Step 6: Cell C2 is a 1

Explanation:

In Column C, we have a 3/ clue that can be solved by (1,3), (2,6), or (3,9) and a 42x clue that can be solved by (1,6,7), or (2,3,7). Consider the possibilities for the 3/ clue. It is impossible for the 3/ group to be solved by (2,6) because 2 eliminates the (2,3,7) possibility for the 42x and 6 eliminates the (1,6,7) possibility. This is guaranteed because the 7 occupies cell B3, rather than a cell in Column C. Since the 3/ cannot be (2,6), it must be (1,3) or (3,9). Both these options guarantee that a 3 solves either cell C6 or cell C7. Either way, Column C's 3 will be in the 3/ group. Therefore, the 42x group cannot be solved by (2,3,7), so it must be solved by (1,6,7). With a 1 in cell I3, the 1 in the 42x clue must solve cell C2.

Follow-up:

With the 1 in cell C2, 6 must solve cell C3. Furthermore, the 28x clue must be solved with a 4 in cell C4 and a 1 in cell D4. The 1 in cell C2 also forces the 3/ group to be solved by (3,9).



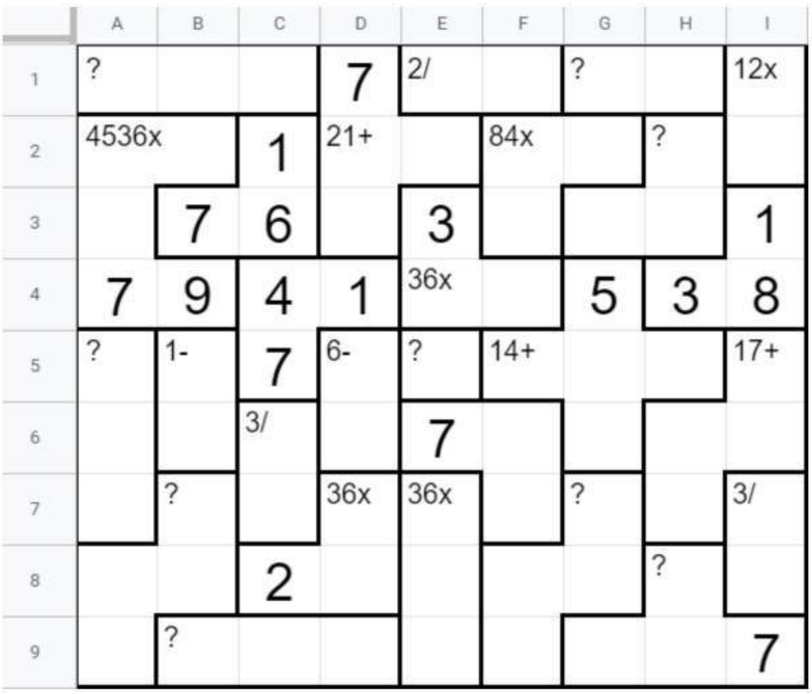
Step 7: Cell C8 is a 2

Explanation:

In Column C, we have found the positions of the 1, 4, 6, and 7 with certainty, and we know the 3/ group is solved by (3,9). Out of the remaining digits, 2, 5, 8, only the first is a factor of 36, so 2 must solve cell C8.

Follow-up:

The five options for a 36x group are (1,4,9), (1,6,6), (2,2,9), (2,3,6), or (3,3,4). We can reduce the options to (2,2,9) and (2,3,6) because we have guaranteed at least one 2.



Step 8: Cell D9 is a 9

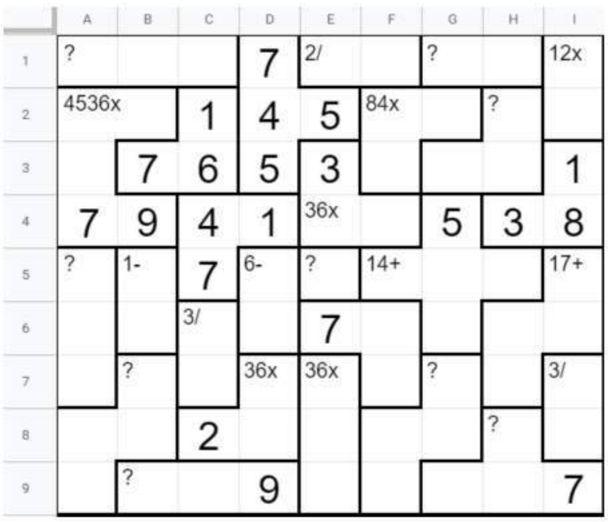
Explanation:

A 6- group can be solved by (1,7), (2,8), or (3,9). The 6- clue in Column D cannot be solved by (1,7) because both digits already appear in Column D. The 6- group also cannot be solved by (3,9). This is because the five options for a 36x group are (1,4,9), (1,6,6), (2,2,9), (2,3,6), and (3,3,4). Because a 2 solves cell C8, we can eliminate the choices down to (2,2,9) and (2,3,6). If the 6- clue was solved by (3,9), then both of these remaining 36x options would be eliminated. Therefore, the 6- clue is solved by (2,8). Since the 2 in Column D is used in the 6- clue, the 36x group must be solved by (2,3,6).

There are three remaining digits of which we do not know the relative positions in Column D: 4, 5, and 9. Notice that two of these numbers will solve the 21+ clue along with a 7. Suppose these two numbers are 5 and 9. This case is impossible because these three numbers would already add to 21, and there is still one more digit in cell E2 to be in the sum. Similarly, these two numbers cannot be 4 and 9 because this would force cell E2 to be a 1. However, the 1 in Column E is in the vertical 36x clue. Therefore, the two numbers must be 4 and 5. This leaves the 9 to solve cell D9.

Follow-up:

We see that 4 and 5 solve cells D2 and D3. Since the sum of 4,5, and 7 is 16, this forces cell E2 to be solved by a 5. With a 5 in cell E2, cell D2 cannot be a 5, so it must be a 4. Then cell D3 is a 5.



Step 9: Cell H2 is a 8

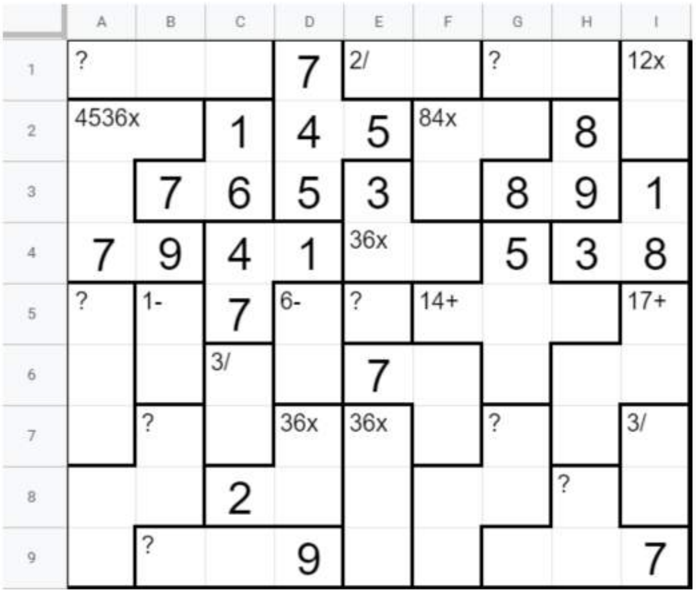
Explanation:

In Row 2, the 8 cannot be in the 12x or 84x groups because 8 is a factor of neither. After dividing 4536 by 7 and then by 9, we are left with 72. Although 8 is a factor of 72, it cannot be in the 4536x group either. If we have an 8 in the 4536x clue, the remaining two digits have to multiply to 72/8 or 9, giving the two possibilities of (1,8,9) and (3,3,8). The former is impossible because two 1s already appear in Rows 2 and 3, and the latter is impossible because this would force a 3 into cell A3 and there is already a 3 in cell E3. Therefore an 8 cannot appear in the 4536x clue. The only remaining option for the 8 in Row 2 is cell H2.

Follow-up:

Using similar logic for the 8 in Row 3, the 8 is limited to cell G3 or cell H3. With an 8 in cell H2, the 8 in Row 3 must solve cell G3.

We can also show the 9 in Row 3 solves cell H3 because it is not a factor of 84. If the 9 in Row 3 solved cell A3, then the product of cells A2 and B2 must be 8, giving two options: (1,8) and (2,4), both of which are impossible because a 1 and 8 already have been deduced in Row 2 in cells C2 and H2, respectively. Therefore, cell H3 is solved by a 9.



Step 10: Cell A3 is a 4

Explanation:

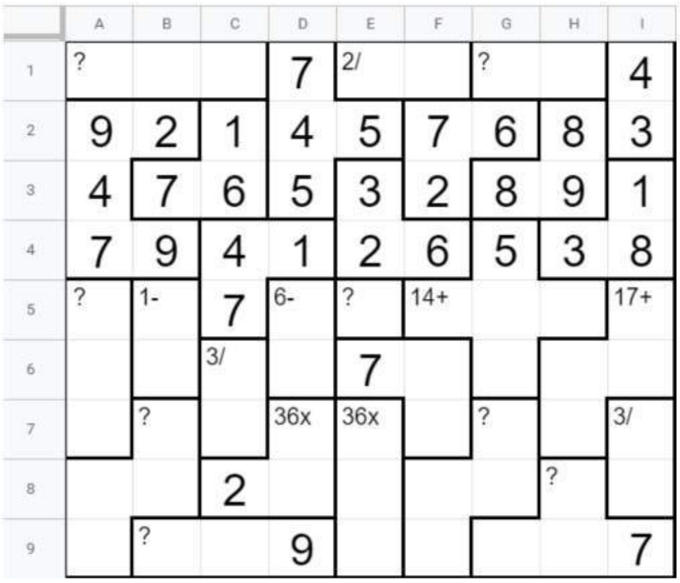
In Row 3 we have only the 2 and 4 remaining. If 2 solved cell A3, then the product of cells A2 and B2 must be 36, giving one possibility: (4,9). However, the 4 in Row 2 already solves cell D2, so it cannot be in the 4536x group and thus cell A3 cannot be 2. Therefore, cell A3 is a 4.

Follow-up:

With a 4 in cell A3, the 2 must solve cell F3. Then the product of cells F2 and G2 is 42, so they are 6 and 7 in some order. Also, 2 cannot solve cell F4 so 6 must solve it. Then, 2 must solve cell E4. Since 6 solves cell F4, 7 must solve cell F2, and 6 must solve cell G2.

Furthermore, because 4 solves cell A3, the product of cells A2 and B2 is 18. With a 6 in either cell F2 or cell G2, it cannot be in either cell A2 or cell B2. The only other option to multiply to 18 other than (3,6) is (2,9). The 9 in cell B4 forces the 9 to solve cell A2, leaving cell B2 to be solved by a 2.

The last digit in Row 2 is a 3, which goes in cell I2. This makes cell I1 to be solved by a 4.



Step 11: Cell E5 is a 8

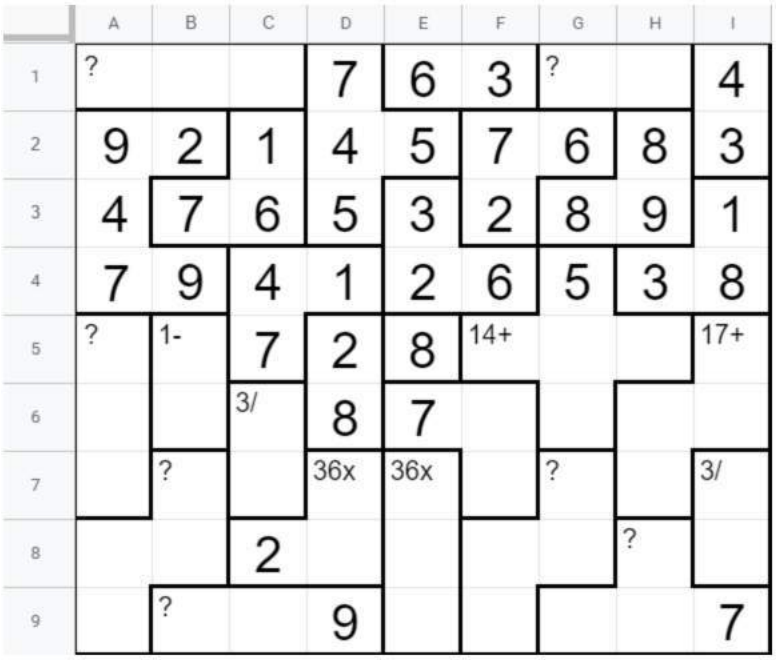
Explanation:

In Column E, the 36x group is solved by (1,4,9). Then the digits 6 and 8 must be in cells E1 and E5. If 8 were to solve cell E1, then cell F1 needs to be solved by a 4. There is already a 4 in cell I1, so the 8 cannot be in cell E1. Therefore, the 8 in Column E solves cell E5.

Follow-up:

Clearly, the 6 in Column E has to solve cell E1 as it is the only choice left. Then cell F1 needs to be solved by a 3 because it needs to satisfy the 2/ clue given.

The 6- clue in Column D is solved by (2,8). Since 8 solves cell E5, then the 2 must solve cell D5, and the 8 solves cell D6.



Step 12: Cell H5 is a 4

Explanation:

In Column I, we are missing the 5 and 9, which are to solve cells I5 and I6. They are both in the 17+ group. Since 5 and 9 sum to 14, the other two digits must sum to 3. The only way to solve this is 1 and 2.

In the 14+ group, we have already deduced that cell G4 is solved by a 5. The other four cells must add to 9. In Column H, we already have solved for the relative positions of the 1, 2, and 3. The 1 and 2 will be in cells H6 and H7, and the 3 solves cell H4. The next smallest choice is a 4. If 4 solves cell H5, then the other three add to 5, which has two options: (1,1,3) and (1,2,2). Cell H5 can also be solved by a 5, which has one option: (1,1,2). This would force cell G5 to be solved by a 2, when there is already a 2 in cell D5. Therefore, cell H5 is not solved by a 5. If cell H5 were solved by a 6, the other three digits add to 3, which is impossible. Therefore cell H5 is solved by a 4.

Follow-up:

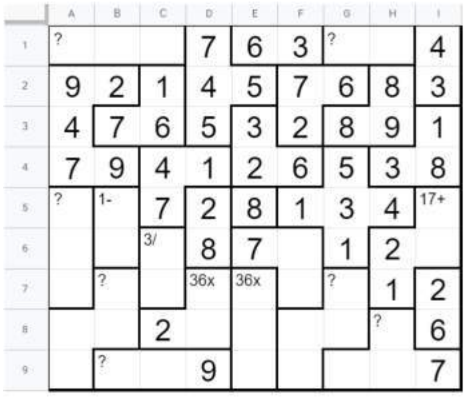
In fact, we know the other three digits in the 14+ cell are (1,1,3) because we cannot have the (1,2,2) option. This means 3 must solve cell G5, and 1 must solve both cells F5 and G6.

Another way to show that cell H5 cannot be solved by a 5, we can fill in some more numbers on the right. In the 3/ group in Column I, because the 3 solves cell I2, the 3/ must be solved by (2,6). With a 2 in cell C8, the 2 in the 3/ group must be in cell I7, and 6 must solve cell I8. In the 17+ clue, we know that cells H6 and H7 are solved by 1 and 2. Since there is a 2 in cell I7, the 2 in the 17+ must appear in cell H6, and then 1 must be in cell H7. The 2 in cell H6 will lead to the 2 in cell D5, and we can make the same argument as above.

Step 13: Cell I5 is a 9

Explanation:

In Row 5, there are only the 5, 6, and 9 not solved. There are 9s in cells A2 and B4, so the 9 cannot be in cell A5 or cell B5. Therefore, the 9 in Row 5 is forced to be in cell I5.



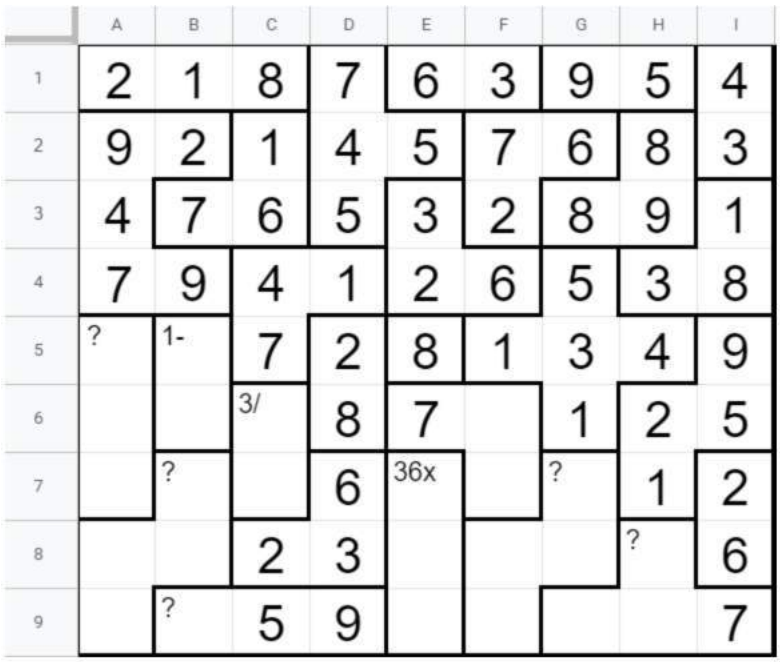
Follow-up:

With the 9 in cell I5, cell I6 must be solved by a 5.

On an unrelated note, because a 6 solves cell I8, cell D8 cannot be solved by a 6. Therefore, the 3 must solve cell D8, and the 6 must solve cell D7.

We can also use the rule that one copy of each digit is allowed per row and per column to deduce that cell H1 is solved by a 5, cell G1 by a 9 (because there is a 9 in the 3/ group in Column C), cell A1 by a 2, cell B1 by a 1, and cell C1 by a 8.

In Column C, the only cell left for a 5 is in cell C9.



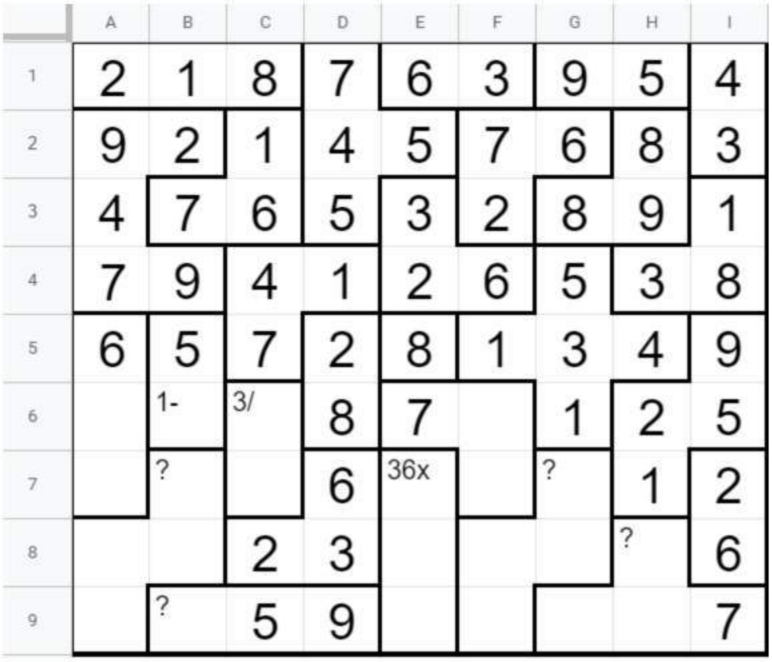
Step 14: Cell B5 is a 5

Explanation:

In Row 5, we are only missing the 5 and 6. Although we have no 5s or 6s to directly eliminate either choice, if we consider what combination of digits could solve the 1- clue, with 1, 2, 7, and 9 already occupying cells in Column B, the only choices left are (3,4), (4,5), and (5,6). If Cell B5 is solved by a 6, this would force a 5 to solve cell B6, but there is already a 5 in Row 6 in cell I6. This case is impossible, so cell B5 is not solved by a 6. Therefore, cell B5 is solved by a 5.

Follow-up:

Cell A5 is solved by a 6.



Step 15: Cell B6 is a 6

Explanation:

In Row 6, we are missing a 6. The only empty cell without a 6 in its column is cell B6.

Follow-up:

The following can be solved for by following the rule of one use of each digit per row and per column:

Cell H8 is solved by a 7, and cell H9 by a 6.
Cell G7 is also solved by a 7, cell G9 by a 2, and cell G8 by a 4.
Cell F6 is solved by a 4, cell C6 by a 9, cell A6 by a 3, and cell C7 by a 3.
Cell B9 is solved by a 3, cell B7 by a 4, and cell B8 by a 8.
Cell E9 is solved by a 4, cell E8 by a 1, cell E7 by a 9.
Cell F8 is solved by a 9, cell F7 by a 5, cell F9 by a 8.
Cell A7 is solved by an 8, cell A8 by a 5, and cell A9 by a 1.

And we arrive at the finished grid below:

	A	B	C	D	E	F	G	H	I
1	2	1	8	7	6	3	9	5	4
2	9	2	1	4	5	7	6	8	3
3	4	7	6	5	3	2	8	9	1
4	7	9	4	1	2	6	5	3	8
5	6	5	7	2	8	1	3	4	9
6	3	6	9	8	7	4	1	2	5
7	8	4	3	6	9	5	7	1	2
8	5	8	2	3	1	9	4	7	6
9	1	3	5	9	4	8	2	6	7

Verbal Arithmetic

Nathan Kuo

A math teacher and an English teacher walk into a bar. As they chat about the weather and New York sports teams, they accidentally exchange homework. When the math teacher's students get their homework, they notice that there are far too many letters on the paper:

Problem 1:

$$\begin{array}{rr} & \text{ABC} \\ \times & \text{DE} \\ \hline & \text{HGBC} \end{array}$$

A, B, C, D, E, H, G are non-zero numbers and less than 8.

Problem 2:

$$\begin{array}{rr} & \text{BOB} \\ \times & \text{BOB} \\ \hline & \text{MARLEY} \end{array}$$

B, O, A, R, L, E, Y are less than or equal to 9.

Problem 3:

$$\begin{array}{rr} & \text{AA} \\ \times & \text{AA} \\ \hline & \text{ABA} \end{array}$$

A, B are non-zero numbers less than 3.

Problem 4:

HI $\div O = I$

H, I, O, are non-zero numbers less than 7.

Problem 5:

AYE $\div NAY = A$

A, Y, E, N are non-zero numbers.

Problem 6:

$$\begin{array}{rr} & \text{abc} \\ + & \text{bc} \\ \hline & \text{acd} \end{array}$$

a, b, c, d are non-zero numbers less than 7.

Solutions to Verbal Arithmetic

Nathan Kuo

Problem 1:

A = 1, B = 2, C = 5, D = 3, E = 7, H = 4, G = 6

Problem 2:

B = 3, O = 5, Y = 9, E = 0, M = 1, L = 6, R = 4, A = 2

Problem 3:

A = 1, B = 2

Problem 4:

H = 2, I = 4, O = 6

Problem 5:

A = 2, Y = 4, E = 8, N = 1

Problem 6:

a = 1, b = 3, c = 2, d = 6, e = 4

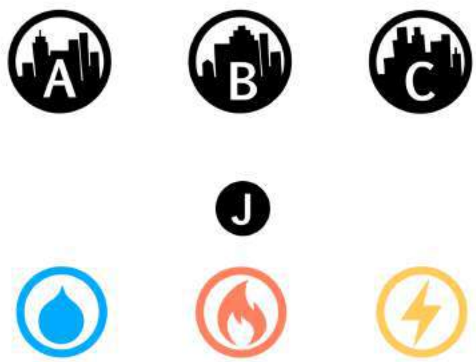
Utilities

Nathan Kuo

Three cities are supplied water, gas, and electricity from three separate plants. Pipes need to be drawn from each supplier to each city. The lines cannot cross more than once. These puzzles draw inspiration from the Three Utilities Puzzle.

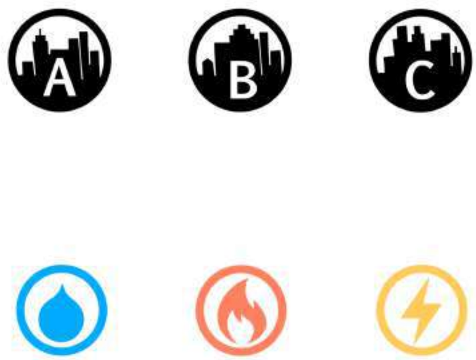
Puzzle 1:

A junction exists between the cities and utility plants. You've been given special pipes that can bend in any way, so curved paths are possible.



Puzzle 2:

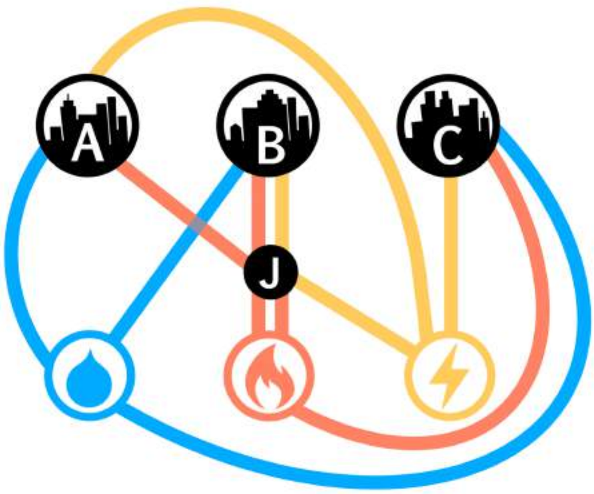
City A needs a double supply of water, while City B doesn't need any water. Unlike the first puzzle, you only have pipes that are either straight, 90 degrees, or 45 degrees.



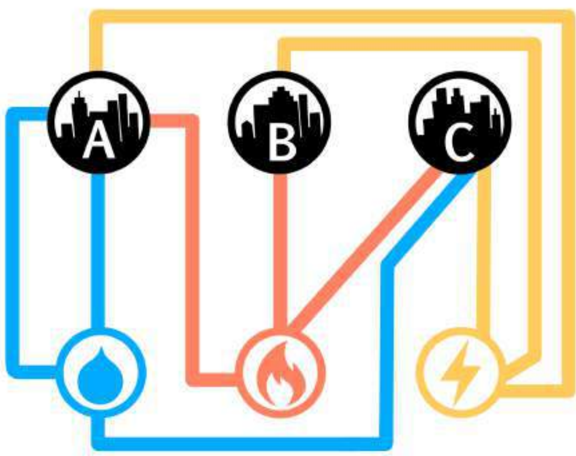
Solutions to Utilities

Nathan Kuo

Solution to Puzzle 1:



Solution to Puzzle 2:



Math Square

Andrew Juang

Puzzle:

	+	21	+		-		-		10
-		-		+		÷		+	
9	-		×		-		+		-287
+		×		+		-		-	
	÷		-		-		+		-11
+		-		×		-		+	
	-	17	+		+		×		39
-		-		+		+		×	
	+		-		÷	4	+		38
3		-48		163		-17		103	

Instructions:

- Fill in the missing numbers.
- The missing values are the whole numbers between 1 and 25.
- Each number is only used once.
- Each row is a math equation.
- Each column is a math equation.
- Remember that the order of operations, i.e., PEMDAS, still applies.

Solution to Math Square

Andrew Juang

7	+	21	+	15	-	10	-	23	10
-		-		+		÷		+	
9	-	19	×	16	-	5	+	13	-287
+		×		+		-		-	
18	÷	2	-	6	-	22	+	8	-11
+		-		×		-		+	
11	-	17	+	20	+	1	×	25	39
-		-		+		+		×	
24	+	14	-	12	÷	4	+	3	38
3		-48		163		-17		103	

Questions and Answers with Distinguished Guest Dr. Lisa Su, President and CEO of Advanced Micro Devices, Inc. (AMD)

Dr. Lisa T. Su is AMD President and Chief Executive Officer, a position she has held since October 2014, and serves on the AMD Board of Directors.

Prior to joining AMD, Dr. Su served as senior vice president and general manager, Networking and Multimedia, at Freescale Semiconductor, Inc. (a semiconductor manufacturing company), and was responsible for global strategy, marketing, and engineering for the company's embedded communications and applications processor business.

From 1994 to 2007, Dr. Su spent 13 years at IBM in various engineering and business leadership positions, including vice president of the Semiconductor Research and Development Center responsible for the strategic direction of IBM's silicon technologies, joint development alliances and semiconductor RD operations.

Dr. Su earned bachelor's, master's and doctorate degrees in electrical engineering from the Massachusetts Institute of Technology (MIT). She has published more than 40 technical articles and was named a Fellow of the Institute of Electronics and Electrical Engineers in 2009. In 2018, Dr. Su was elected to the National Academy of Engineering and received the Global Semiconductor Association's Dr. Morris Chang Exemplary Leadership Award. In 2019, Dr. Su was named one of the World's 50 Best CEOs by Barron's, one of the "Most Powerful Women in Business" by Fortune Magazine, and she was included in "The Bloomberg 50," a list of people who defined the year. In 2020, Fortune named Dr. Su #2 on its "Business Person of the Year" list, she was *elected to the American Academy of Arts Science, received the Grace Hopper Technical Leadership Abie Award, and was honored by the Semiconductor Industry Association with its highest honor, the Robert N. Noyce Award. Dr. Su has been a member of the board of directors of Cisco Systems, Inc., since January 2020. She also serves on the board of directors for the Semiconductor Industry Association. (Advanced Micro Devices)*

Dr. Su's answers are taken from an interview conducted by Josephine Lee and Mr. Kevin Anderson from AMD and have been edited for clarity.

1) Who were your great mentors when you were a "kid" and how did they help you?

My mom's been a large inspiration in my life – my parents' sacrifice and commitment to excellence were key drivers for me. The pushed me to realize my full potential. And they never expected anything from me that they didn't expect from themselves.

2) What building-block experiences should teenagers who are interested in STEM look for?

I like to believe that my training as an engineer is about how to solve problems. How can you analytically solve a problem and make that fun? And that is one of the key parts of making STEM interesting, especially to women and underrepresented minorities.

3) How did growing up in NYC influence your love for science and technology and your outlook on life?

As a child, I spent time taking apart my brother's broken remote-control cars and repairing them. That curiosity and determination to fix an issue helped pave my way to be interested in engineering and technology.

In addition, I was very fortunate to have attended the Bronx High School of Science. Bronx Science students take a college preparatory curriculum that includes four years of lab science, math, English, and social studies. We also had the opportunity to do independent research.

When I was in college, the hottest majors were electrical engineering and computer science. Electrical engineering dealt with hardware, and I wanted to actually build things. Most people also said it was the "harder" major and I thought I would definitely learn a lot. I learned when I chose something very difficult and did well, it would give me great confidence for the next challenge.

4) What advice would you give your 16-year-old self?

One, a good, strong education is very important. Nobody can take your education away from you; it is your calling card. Two, be ambitious in what you do. And, three, take feedback to heart and learn from your failures. I really believe that's what makes you better.

I was given a piece of advice when I was a young engineer at IBM that has really stuck with me. Someone told me, "You should run toward problems." If you're going to work on something, work on something that's really important.

5) Why do you think women are underrepresented in STEM-related fields? How have you tried to address this gap?

Exposure to STEM subjects is extremely important to young people. There are incredible opportunities for women and minorities. It is also important that you can see the fruits of your work and their impact on society.

At AMD, we're especially passionate about enabling the imagination and creativity of the next generation. Technology in their hands encourages exploration, imagination and learning – opening doors to new careers and possibilities. That's why we're partnering with local organizations in four cities to establish AMD-powered learning labs to inspire students to pursue STEM education. AMD is providing hardware and other support in each new lab, as well as ongoing engagement from our employee volunteers. These four AMD Learning Labs are expected to impact more than 1,000 students.

Education provides incredible opportunity for students to expand their horizons and achieve their fullest potential. And with access to powerful technology in the classroom, they may be the future innovators, researchers, leaders or entrepreneurs to dream up new ways to solve society's challenges.

6) As an executive who has to make long-term decisions today, how should students learn the skill of making long-term decisions?

When I took over as CEO at AMD, I felt that what we needed to do was focus on our core competency, which for AMD is high-performance computing. This meant setting a 5-year vision and roadmap for how we would build the best products in the industry. We laid out a clear strategy based on focus and disciplined execution and asked the whole organization to align around them. This also meant deciding not to do some things, like not pursuing some markets that were good markets, but just not in our core competency. Deciding what not to do is as important as deciding what to do.

7) Many of today's engineers choose software over hardware. What are the advantages of choosing a career in hardware?

A software and hardware career can both be very rewarding for the right person. In hardware, you get to "take things apart," examine, fix, and put it back together again.

It is one of the fields where you can make a difference by the choices you make on the technology front.

8) How do you see the semiconductor industry impacting our lives five years from now?

This is a very unique time in the semiconductor market in which I think people are just realizing how important chips are to every part of our lives. That's why I am in this business. It's really humbling to know that the chips we are building are so important no matter who you are and what you are trying to do.

In technology, particularly in semiconductors, we have to make bets far in advance. Decisions made today will impact products that come to market three or four years from now. We do spend a lot of time with our customers learning what problems they are facing so that we can develop new solutions. We also spend time studying overall technology trends, identifying the big bottlenecks, and key innovations that are on the horizon.

Questions and Answers with Mr. Bram Cohen, Class of 1993

Mr. Bram Cohen, Class of 1993, famously created the peer-to-peer BitTorrent protocol in 2005 in his early thirties and the Chia cryptocurrency, the first enterprise-grade digital money, in 2017 in his forties. As a Stuyvesant student, he participated in Math Team and qualified for the USA Math Olympiad. During his high school years, Cohen also attended the Hampshire College Summer Studies in Mathematics program. Although he attended SUNY Buffalo for one year, he ultimately left to work for several startup companies, such as Valve. Then, he founded BitTorrent, which grew to fame, with over 150 million global users, due to its ability to quickly share large music and movie files online. Later, cryptocurrency startup TRON acquired BitTorrent, and Cohen left to start the groundbreaking Chia Network, a proof of space-time cryptocurrency. Chia launched in May 2021 and uses about 10,000 times less energy than what is usually needed to mine proof-of-work-based cryptocurrencies such as Bitcoin. Its self-reported market cap is currently at around $244 million and has driven hard drive disk sales soar to record levels, increasing by 141 percent.

Mr. Bram Cohen's answers are taken from an interview conducted by Josephine Lee and have been edited for clarity.

1) How did you get interested in math and computer science? Did you always envision being an entrepreneur?

Reading books by Martin Gardner, when I was a kid, actually was a lot of my introduction to math. There were also collections of his mathematical games and columns from *Scientific American* that I read rather obsessively. Then I was part of Math Team starting in junior high. As far as being an entrepreneur, I didn't decide, "Oh, I want to be an entrepreneur." I decided, "Hey, there's this project I want to work on." And that turned into being an entrepreneur. So I guess I was always an entrepreneur.

2) What do you remember most about your experience at Stuyvesant?

Most of my experience at Stuyvesant was really centered around being on Math Team, which was run by Mr. Geller at the time. Mostly, I just remember that Math Team was actually fun.

3) How did you get BitTorrent started? What was the process in creating BitTorrent and getting people to use it? How were you able to get your technology widely adopted?

Oh, well, I just kind of wrote it. I had worked at a bunch of startups doing protocol stuff. And from those experiences, I had thoughts of something that would be useful. So I built the thing that was useful, and then I released it, and it was free, and people found it helpful. So they used it.

4) What was the process in creating Chia? Are there any difficulties in expanding, creating, and distributing Chia that you hadn't experienced before?

Chia is a much bigger project than BitTorrent in terms of scope, getting widespread mass adoption and enabling real commerce in the world. I come into Chia having a much, much better idea of what I'm doing. I also have more of a network, so that's been better. But it's also been a much larger project to even get started in the first place.

5) What advice do you have for high school students who want to have a career in CS and/or to be an entrepreneur in CS? Are there particular skills one should focus on?

For being an entrepreneur, I would say the big mistake many people make is jumping directly into being an entrepreneur with no work experience. You're way, way better off working at startups, where you're one of the first ten employees or something like that. Because then, you get some idea what's going on in it. You build a network, and you have some idea what to expect rather than just flailing completely on your own. It's better to be paid to gain an audience than just to give away money. And then, after you have some experience, then you can try being an entrepreneur.

6) What lessons have you learned by going through the process of developing, marketing, and selling a company that might be helpful for high school students who want to be entrepreneurs?

Chia is very tech-heavy and has a firm focus there since we're trying to be the next chapter of digital money. The weakness that most companies have is in the product, not so much in engineering. People have reasonably good ideas on how to do engineering, for the most part, these days. But the whole way in which companies approach, "Okay, what are we doing? Why are we doing it?" tends to be very slapdash and haphazard. Partially this isn't done well because there aren't obvious guidelines about how to get better at it, but as an entrepreneur, you should try to focus on really understanding what things are like from a product standpoint.

Stuyvesant Faculty Perpectives:
Mrs. Deena Avigdor:
Seeing the Beauty of Mathematics and New York City

Debolina Kunda, Josephine Lee

Mrs. Deena Avigdor has guided students through mathematics at Stuyvesant since 1996. She earned her Bachelor's degree in mathematics from the University of California, Los Angeles, and a Master's degree in Science and Teaching from the University of New Hampshire at the start of her teaching career. Over the past 25 years at Stuyvesant, she has taught the full spectrum of math classes, including Geometry, Precalculus, Calculus Applications, AP Calculus AB, AP Calculus BC, Multivariable Calculus, and Linear Algebra. Currently, she primarily teaches Geometry and AP Calculus BC.

This article is based on an interview conducted by Josephine Lee. The content has been edited for clarity.

1 Learning Math, Teaching Math

From pre-med to math at UCLA—how did this happen? The simple act of sitting in her room reading her Anton Calculus textbook led to Deena Avigdor's road to pure mathematics. Riemann sums in particular caught her attention.

The abstraction in all forms of mathematics, from Geometry to Algebra, intrigued Avigdor from the beginning. In her opinion, the most fascinating aspect of math is that "you could take something from the physical world and abstract it, and it also works in the reverse. You could take the prime number theorem, something this abstract, and then find applications for it. It kind of works both ways."

Teaching cemented her career in math. At the parochial high school where Avigdor had attended, her former Calculus teacher broke his hip and a Geometry teacher took maternity leave. These events allowed Avigdor to teach in their place and to realize that math and teaching were her callings.

2 Teaching in Stuyvesant Then

Over the past quarter-century, Avigdor has taught a wide variety of classes, from Geometry to Linear Algebra. When Avigdor first started teaching Geometry, the first math course taken by a typical Stuyvesant student, a former student viewed it as a "downgrade" in course levels for her. However, Avigdor believes that teaching Geometry is just as challenging and rewarding as teaching Calculus, so she thoroughly enjoys teaching both.

Some may have thought math teachers have significantly less work than those in other departments since "[they] don't have to grade essays." However, Avigdor explains that "the challenge of dealing with homework in a meaningful way on a daily basis is on par with what other people are doing." She has found that grading a math test is quite challenging because she has to precisely determine where students went wrong to accurately give them partial credit. There is a significant "gray area" in partial credit as there are various types of mistakes. Avigdor has always endeavored to provide meaningful feedback on every assignment and test for over 150 students, which is quite a feat.

Avigdor recognizes that another challenging aspect of teaching math has been emphasizing and fostering the creativity required to solve complex problems. Although students studying Geometry have to memorize theorems and definitions, these serve as the tools necessary to construct a unique solution. The drawback of high school math is that the curriculum and material are tightly organized in a particular order, but not in the way the material was originally developed. Through proofs and problems with multiple solutions, Avigdor has helped students see the creative aspect of how mathematical tenets came to be.

Avigdor appreciates the unique drive of Stuyvesant students to learn and values how her colleagues provide an environment that has pushed her to grow, whether in math or pedagogy. Three motivators for Avigdor when teaching have been the interaction with the students, the fresh start every school year, and the evolving curriculum. Furthermore, Avigdor has enjoyed teaching students as freshmen and seeing their growth again as seniors. She remarks, "Sometimes it's almost not like the same student!" She has also loved the daily joys of a student asking an insightful question or figuring out another solution to a problem. According to Avigdor, introducing proofs has been one of the most challenging parts of teaching because "you are trying to enable the other person to be able to make the connection without you giving it to them."

3 Teaching at Stuyvesant Now

Even though students are as interested in learning as ever, Avigdor finds that many aspects of teaching at Stuyvesant have changed over time, including the curriculum. For example, much more technology is used now, like Smartboards. In addition, the student population has grown, but the number of teachers has mostly remained the same. Seeing classes grow from 25 students to 34 means having "another row in the back of the classroom." While a larger number of students means a larger pool of ideas, it also means that teachers cannot give enough individual attention to each person. Avigdor believes a class of 22 to 25 students is most optimal.

When the COVID-19 pandemic prompted remote learning, students collaborated through breakout rooms which were very beneficial. However, the virtual setting still lacked the "buzz" or the "certain energy from being in the classroom around other groups" which normally increases productivity. This year, Avigdor used many platforms such as Jamboard, Google Slides, Classkick, and Desmos, which helped her give feedback to the students. She hopes to integrate the technologies and techniques she's learned into her in-person classes in the future. Despite her adaptation to remote instruction, she still found this year to be the most challenging of her career, even surpassing the school year of the 9/11 attacks.

Remote learning also introduced varied interaction between students. For example, Avigdor noticed that when the freshmen this year were put in groups, they interacted with each other by typing instead of speaking. This was not the case as much with her senior classes because they knew each other already and had previous experience doing group work in Stuyvesant classes. In addition, Avigdor opted to type comments instead of visiting individual breakout rooms. In the years before the pandemic, she provided directed notes for her Geometry class since it was unnecessary for them to copy definitions off of the board. They instead only had to fill in the blanks in the directed notes. Her Calculus students did not get directed notes, however. For the freshmen this year, she had a "flipped classroom": Avigdor made videos that they watched for homework to learn the new material, and in class, they solved problems. On the other hand, seniors this year received directed notes since they needed to cover more material in less time than in a typical school year.

4 Beyond the Classroom

So what is the key to a successful career in math? Avigdor advises that it is perseverance. When students train for standardized tests, they tackle short problems and receive immediate answers. However, it takes much longer to find the solution in a mathematical or problem-solving field, so students must learn to stick with it, which is easier said than done.

Impressively, Avigdor has another passion in addition to math: photography. Her interest in photography began back in seventh or eighth grade, and she now immensely enjoys creating photo books with her own layouts and pictures. Interestingly, it was Stuyvesant students enrolled in school photography classes who encouraged her to invest in a sophisticated camera. This summer, Avigdor is taking two photography classes while she continues to work on a project she started about eight years ago. She is determined to capture the beauty of New York City by photographing flowers, which isn't typically associated with New York City, in the Brooklyn Botanical Garden and locally in her neighborhood. Given Avigdor's enthusiastic and dedicated attitude, she will be able to add this project to her photo book library in the near future.

Stuyvesant Math and Computer Science Achievements

2020-2021

American Regions Mathematics League (ARML) Competition
NYC Math Team's top team, Tin Man, finished in 2nd place overall, its best performance in 13 years. Twelve of the fifteen students on the team were Stuyvesant students: Alice Zhu, Benjamin Kreiswirth, Declan Stacy, Ethan Joo, Jerry Liang, Mario Tutuncu-Macias, Max Vaysburd, Maxwell Pang, Paul Gutkovich, Rishabh Das, Srinath Mahankali, and Theo Schiminovich. Co-captain Rishabh Das earned a perfect score on the individual round and finished in 3rd place in the competition.

American Computer Science League (ACSL) competition
Alex Cho won a gold medal with a perfect score in the final round.

USA Junior Mathematical Olympiad (USAJMO)/USA Mathematical Olympiad (USAMO)
Paul Gutkovich was named a USAJMO Winner, finishing among the top 12 students in the country. He was selected to attend the Math Olympiad Program (MOP), which is used to select the US International Math Olympiad Team. In addition, Ethan Joo earned an Honorable Mention on the USAMO, scoring among the top 15 students on the exam.

USAMO Qualifiers
Chen, Carol (11th)
Das, Rishabh (11th)
Joo, Ethan (12th)
Mahankali, Srinath (12th)
Mao, Madelyn (12th)
Pang, Maxwell (12th)
Schiminovich, Theo (12th)

USAJMO Qualifiers
Gutkovich, Paul (10th)
Gupta-She, John (10th)
He, Zhengxi (10th)

University of Chicago Outstanding Educator
Mr. Stan Kats, math teacher, was selected as a University of Chicago Outstanding Educator. Three students recently accepted by U Chicago recommended him for this award: Benjamin Gallai, David Hu, and XianJun An.

Princeton University Mathematics Competition (PUMaC)
NYC Math Team won the competition. The top team, Tin Man, was made up of eight students, five of whom were from Stuyvesant - Srinath Mahankali, Rishabh Das, Ethan Joo, Max Vaysburd, and Theo Schiminovich.

Two of the students, Srinath Mahankali and Rishabh Das, along with Mario Tutuncu-Macias and Jerry Liang, two students from Scarecrow, the second team, qualified for the individual finals. Tin Man was ranked first individually, as well as a team.

Scarecrow finished 12th overall. Both teams placed in the top 10 for the proof round (2nd and 6th places). Scarecrow was made up entirely of Stuyvesant students: Declan Stacy, Ben Kreiswirth, Jerry Liang, Mario Tutuncu-Macias, Maxwell Pang, Alice Zhu, Joseph Othman, and Josiah Moltz.

Harvard-MIT Mathematics Tournament (HMMT)
Tin Man placed 3rd overall and also finished 2nd in the team round and 5th in the guts round. Scarecrow placed 13th overall, 16th in team, and 11th in guts. Rishabh Das, Maxwell Pang, and Ethan Joo placed among the top 50 individuals, earning spots to compete in the Harvard-MIT Invitational Competition (HMIC).

New York State Mathematics League (NYSML)
The NYC Math Team extended its streak of wins at the NYSML competition to probably almost 50 years in a row. Tin Man had a decisive win over Bergen County Academies (competing unofficially because they're from New Jersey) of nearly 30 points. Scarecrow and Jollipop Juild were not far behind. Overall, NYC Teams ranked 1, 3, 4, 7,10,11 out of the 31 teams that competed. Mario Tutuncu-Macias was the individual winner, Alice Zhu placed 4th, and Rishabh Das and Maxwell Pang both scored perfect 10s in the individual portion as well.

Association for Women in Mathematics's Essay Contest, Grades 9-12
Angela Cai won with the essay "Stepping Outside the Mold to Improv-e Mathematics" about Ellen Eischen.

American Mathematics Competition (AMC): Distinguished Honor Roll
Jacob Paltrowitz (10A)
Josiah Moltz (10A)
Kevin Xiao (10B)
Paul Gutkovich (10B)
Ethan Joo (12A)
Rishabh Das (12A and 12B)

94 American Invitational Mathematics Examination (AIME) Qualifiers
Allen Baranov, Angela Cai, Ian Chen Adamczyk, Andy Chen, Carol Chen, Ivan Chen, James Chen, Jonathan Chen, Quentin Chen, Ryan Chen, Lukas Chin, Rishabh Das, Maya Doron-Repa, Konstantin Dubovsky, Lee Eisenberg, Yusuf Elsharaway, Lingjie Fang, Sachin Fosenka, Rin Fukuoka, Benjamin Gallai, Joshua Gao, John Gupta-She, Paul Gutkovich, Jesse Hammer, Cary Han, Zhengxi He, David Hu, Eric Huang, Joseph Jeon, David Jiang, Vincent Jiang, Allison Jin, Ethan Joo, Andrew Juang, Eric Kim, Joseph Kim, Jacob Kirmayer, Julia Kozak, Benjamin Kreiswirth, Milind Latwani, Newton Le, Erin Lee, Josephine Lee, Mary Lee, Ryan Lee, Steven Lee, Alvin Li, Andrew Li, Justin Li, Jerry Liang, Daisy Lin, Ethan Lin, Mikayla Lin, Vincent Lin, William Lin, Alice Liu, Katherine Liu, Haokai Ma, Srinath Mahankali, Madelyn Mao, Nora Miller, Josiah Moltz, Brandon O'Conner, Joseph Othman, Aditya Pahuja, Jacob Paltrowitz, Maxwell Pang, Gitae Park, Wonjong Park, Rayat Roy, Theo Schiminovich, Shadman Shaharia, Jeremy Shen, Declan Stacy, Jennifer Sun, Ruiwen Tang, Tyler Tsang, Mario Tutuncu-Macias, Maxim Vaysburd, Bwohan Wang, Ivan Wei, Chun Yeung Wong, Edward Wu, Kevin Xiao, Diana Yang, Max Yenlee, Derek Zang, Maxwell Zen, Anna Zhang, Bryan Zhang, Calvin Zhang, Kathleen Zhang, Elie Zhang, Alice Zhu

Math Prize for Girls Qualifiers
Angela Cai
Carol Chen
Josephine Lee
Kathleen Zhang
Alice Zhu

Mathematics and Computer Science Department Senior Awards
The Dr. Richard Rothenberg Memorial Award for Highest Honors in Mathematics: Srinath Mahankali

The Richard B. Geller Memorial Scholarship for Mathematics: Ethan Joo, Madelyn Mao

Henrietta O. Midonick Award for Excellence in Mathematics: Anna Zhang

David J. Pichler Memorial Scholarship: Carter Ley

The Okean Family Memorial Award for Scholarship in Pure or Applied Mathematics: Mario Tutuncu-Macias, Theo Schiminovich

Irene Finkel Memorial Award for Excellence in Mathematics: Maxim Vasyburd, Maxwell Pang

Highest Honors in Computer Science: Mario Tutuncu-Macias, Ruoshui Mao

High Honors in Computer Science: Kristof Misquitta, Steven Lee, Eric Lo, Vincent Jiang, Jin Cheng Zhang, Victoria Gao, Michael Nath

Honors in Computer Science: Victor Siu, Graziella Cantarella, Tristan Lee, Liam Kronman, Nehemiah Yu, Renee Mui, Madeleine Andersen, Cindy Zheng, Vladislav Vostrikov

Golden Ducky for Service to the Computer Science Community: Stella Oh, Graziella Cantarella, Madelyn Mao, Abir Taheer, Helena Williams, Amelia Chin, Zoe Piccirillo, Ethan Shenker, Ruoshui Mao, Pak Ming Lau, Maxim Vaysburd, Mario Tutuncu-Macias, Victoria Gao, Jonathan Lee, Carlos Hernandez, Julianna Yu, Dylan Lane

Decades-Old Computer Science Conjecture Solved in Two Pages

Distinguished Guest Contributor
Dr. Erica Klarreich, award-winning journalist

Have you ever doubted your ability to solve a problem no one else has solved before? How can one researcher contribute to a field dominated by world-renowned experts and brilliant innovators? In this article, Dr. Erica Klarreich, award-winning mathematics and science journalist, explores Professor Hao Huang's simple yet elegant proof to a conjecture that went unproven for nearly three decades. Dr. Huang's breakthrough while in the Department of Mathematics at Emory University in 2019 serves as a reminder that with discipline and passion we can all strive to make important contributions. It is also an example of how the intersection of math and computer science can lead to discovery. His proof of the "sensitivity" conjecture now allows electrical engineers to reduce the number of transistors of any given circuit.

This article was reprinted with permission from Mr. Thomas Lin, Editor-in-Chief of Quanta Magazine*, an editorially independent publication of the Simons Foundation. This article can be accessed via: https://www.quantamagazine.org/mathematician-solves-computer-science-conjecture-in-two-pages-20190725/*

A paper posted online this month has settled a nearly 30-year-old conjecture about the structure of the fundamental building blocks of computer circuits. This "sensitivity" conjecture has stumped many of the most prominent computer scientists over the years, yet the new proof is so simple that one researcher summed it up in a single tweet.

"This conjecture has stood as one of the most frustrating and embarrassing open problems in all of combinatorics and theoretical computer science," wrote Scott Aaronson of the University of Texas, Austin, in a blog post. "The list of people who tried to solve it and failed is like a who's who of discrete math and theoretical computer science," he added in an email.

The conjecture concerns Boolean functions, rules for transforming a string of input bits (0s and 1s) into a single output bit. One such rule is to output a 1 provided any of the input bits is 1, and a 0 otherwise; another rule is to output a 0 if the string has an even number of 1s, and a 1 otherwise. Every computer circuit is some combination of Boolean functions, making them "the bricks and mortar of whatever you're doing in computer science," said Rocco Servedio of Columbia University.

Over the years, computer scientists have developed many ways to measure the complexity of a given Boolean function. Each measure captures a different aspect of how the information in the input string determines the output bit. For instance, the "sensitivity" of a Boolean function tracks, roughly speaking, the likelihood that flipping a single input bit will alter the output bit. And "query complexity" calculates how many input bits you have to ask about before you can be sure of the output.

Each measure provides a unique window into the structure of the Boolean function. Yet computer scientists have found that nearly all these measures fit into a unified framework, so that the value of any one of them is a rough gauge for the value of the others. Only one complexity measure didn't seem to fit in: sensitivity.

In 1992, Noam Nisan of the Hebrew University of Jerusalem and Mario Szegedy, now of Rutgers University, conjectured that sensitivity does indeed fit into this framework. But no one could prove it. "This, I would say, probably was the outstanding open question in the study of Boolean functions," Servedio said.

"People wrote long, complicated papers trying to make the tiniest progress," said Ryan O'Donnell of Carnegie Mellon University.

Now Hao Huang, a mathematician at Emory University, has proved the sensitivity conjecture with an ingenious but elementary two-page argument about the combinatorics of points on cubes. "It is just beautiful, like a precious pearl," wrote Claire Mathieu, of the French National Center for Scientific Research, during a Skype interview.

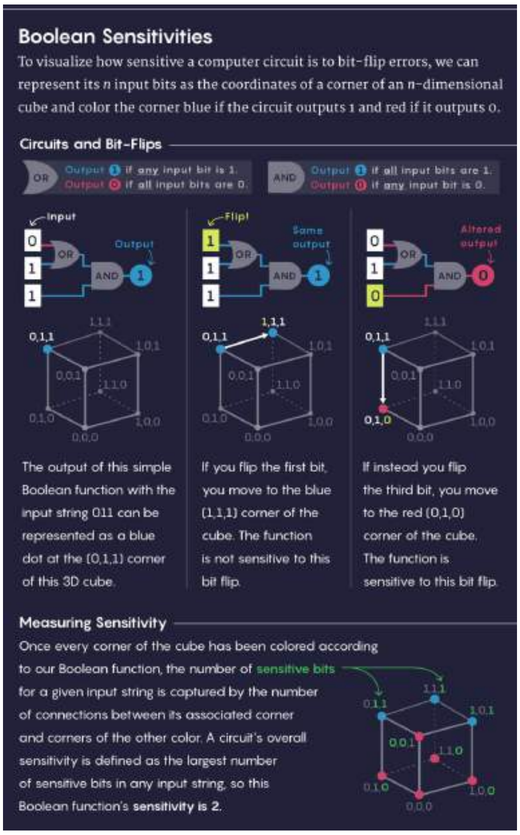
Aaronson and O'Donnell both called Huang's paper the "book" proof of the sensitivity conjecture, referring to Paul Erdős' notion of a celestial book in which God writes the perfect proof of every theorem. "I find it hard to imagine that even God knows how to prove the Sensitivity Conjecture in any simpler way than this," Aaronson wrote.

A Sensitive Matter

Imagine, Mathieu said, that you are filling out a series of yes/no questions on a bank loan application. When you're done, the banker will score your results and tell you whether you qualify for a loan. This process is a Boolean function: Your answers are the input bits, and the banker's decision is the output bit.

If your application gets denied, you might wonder whether you could have changed the outcome by lying on a single question — perhaps, by claiming that you earn more than 50,000 when you really don't. If that lie would have flipped the outcome, computer scientists say that the Boolean function is "sensitive" to the value of that particular bit. If, say, there are seven different lies you could have told that would have each separately flipped the outcome, then for your loan profile, the sensitivity of the Boolean function is seven.

Computer scientists define the overall sensitivity of the Boolean function as the biggest sensitivity value when looking at all the different possible loan profiles. In some sense, this measure calculates how many of the questions are truly important in the most borderline cases – the applications that could most easily have swung the other way if they'd been ever so slightly different.



Lucy Reading-Ikkanda/Quanta Magazine

Sensitivity is usually one of the easiest complexity measures to compute, but it's far from the only illuminating measure. For instance, instead of handing you a paper application, the banker could have interviewed you, starting with a single question and then using your answer to determine what question to ask next. The largest number of questions the banker would ever need to ask before reaching a decision is the Boolean function's query complexity.

This measure arises in a host of settings – for instance, a doctor might want to send a patient for as few tests as possible before reaching a diagnosis, or a machine learning expert might want an algorithm to examine as few features of an object as possible before classifying it. "In a lot of situations – diagnostic situations or learning situations – you're really happy if the underlying rule … has low query complexity," O'Donnell said.

Other measures involve looking for the simplest way to write the Boolean function as a mathematical expression, or calculating how many answers the banker would have to show a boss to prove they had made the right loan decision. There's even a quantum physics version of query complexity in which the banker can ask a "superposition" of several questions at the same time. Figuring out how this measure relates to other complexity measures has helped researchers understand the limitations of quantum algorithms.

With the single exception of sensitivity, computer scientists proved that all these measures are closely linked. Specifically, they have a polynomial relationship to each other − for example, one measure might be roughly the square or cube or square root of another. Only sensitivity stubbornly refused to fit into this neat characterization. Many researchers suspected that it did indeed belong, but they couldn't prove that there were no strange Boolean functions out there whose sensitivity had an exponential rather than polynomial relationship to the other measures, which in this setting would mean that the sensitivity measure is vastly smaller than the other measures.

"This question was a thorn in people's sides for 30 years," Aaronson said.

Cornering the Solution

Huang heard about the sensitivity conjecture in late 2012, over lunch with the mathematician Michael Saks at the Institute for Advanced Study, where Huang was a postdoctoral fellow. He was immediately taken with the conjecture's simplicity and elegance. "Starting from that moment, I became really obsessed with thinking about it," he said.

Huang added the sensitivity conjecture to a "secret list" of problems he was interested in, and whenever he learned about a new mathematical tool, he considered whether it might help. "Every time after I'd publish a new paper, I would always go back to this problem," he said. "Of course, I would give up after a certain amount of time and work on some more realistic problem."

Huang knew, as did the broader research community, that the sensitivity conjecture could be settled if mathematicians could prove an easily stated conjecture about collections of points on cubes of different dimensions. There's a natural way to go from a string of n 0s and 1s to a point on an n-dimensional cube: Simply use the n bits as the coordinates of the point.

For instance, the four two-bit strings − 00, 01, 10 and 11 − correspond to the four corners of a square in the two-dimensional plane: (0,0), (0,1), (1,0) and (1,1). Likewise, the eight three-bit strings correspond to the eight corners of a three-dimensional cube, and so on in higher dimensions. A Boolean function, in turn, can be thought of as a rule for coloring these corners with two different colors (say, red for 0 and blue for 1).

The mathematician Hao Huang during a recent vacation in Lisbon. Photo by Yao Yao

In 1992, Craig Gotsman, now of the New Jersey Institute of Technology, and Nati Linial of Hebrew University figured out that proving the sensitivity conjecture can be reduced to answering a simple question about cubes of different dimensions: If you choose any collection of more than half the corners of a cube and color them red, is there always some red point that is connected to many other red points? (Here, by "connected," we mean that the two points share one of the outer edges of the cube, as opposed to being across a diagonal.)

If your collection contains exactly half the corners of the cube, it's possible that none of them will be connected. For example, among the eight corners of the three-dimensional cube, the four points (0,0,0), (1,1,0), (1,0,1) and (0,1,1) all sit across diagonals from one another. But as soon as more than half the points in a cube of any dimension are colored red, some connections between red points must pop up. The question is: How are these connections distributed? Will there be at least one highly connected point?

In 2013, Huang started thinking that the best route to understanding this question might be through the standard method of representing a network with a matrix that tracks which points are connected and then examining a set of numbers called the matrix's eigenvalues. For five years he kept revisiting this idea, without success. "But at least thinking about it [helped] me quickly fall asleep many nights," he commented on Aaronson's blog post.

Then in 2018, it occurred to Huang to use a 200-year-old piece of mathematics called the Cauchy interlace theorem, which relates a matrix's eigenvalues to those of a submatrix, making it potentially the perfect tool to study the relationship between a cube and a subset of its corners. Huang decided to request a grant from the National Science Foundation to explore this idea further.

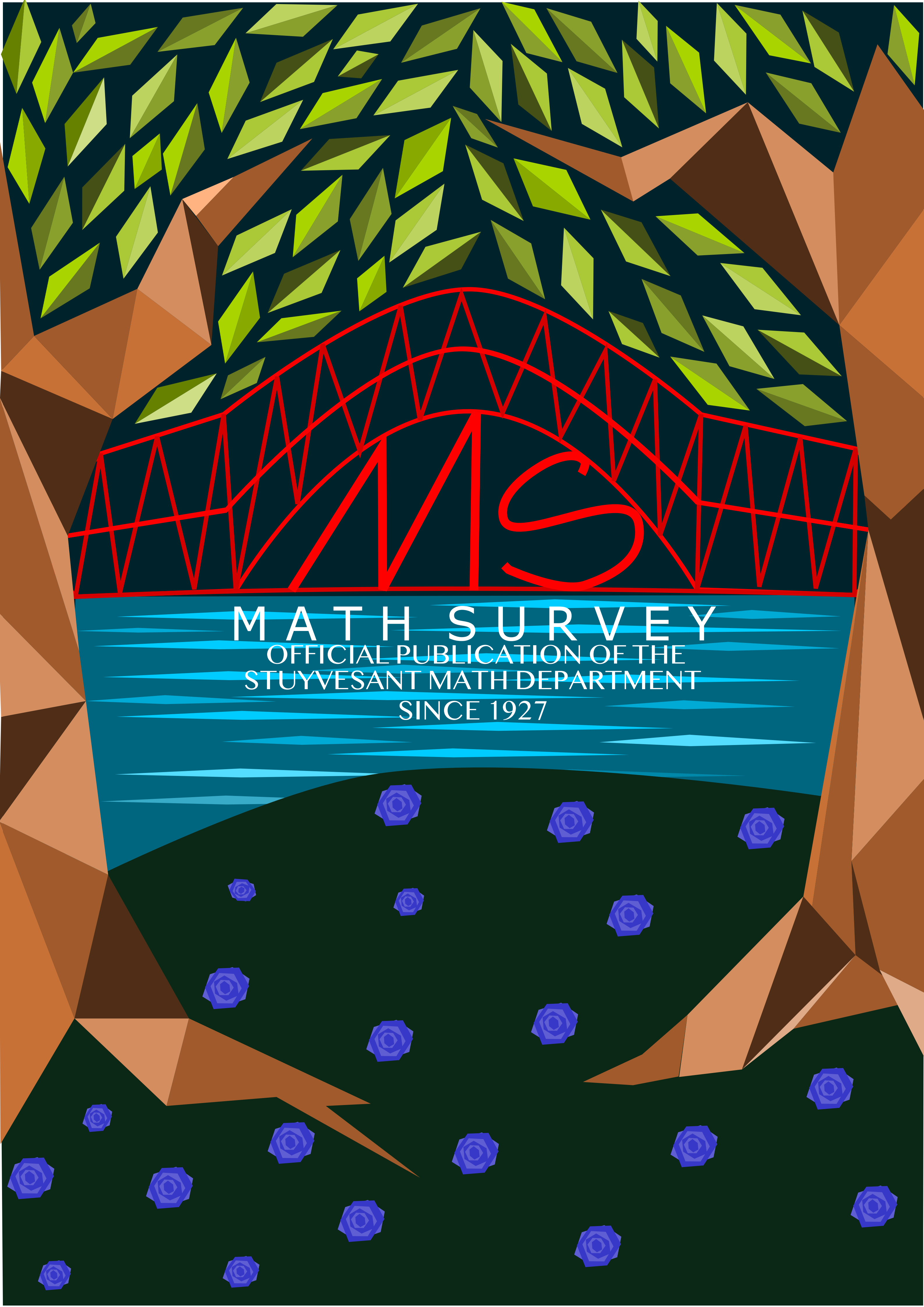
Then last month, as he sat in a Madrid hotel writing his grant proposal, he suddenly realized that he could push this approach all the way to fruition simply by switching the signs of some of the numbers in his matrix. In this way, he was able to prove that in any collection of more than half the points in an n-dimensional cube, there will be some point that is connected to at least $\sqrt{n}$ of the other points – and the sensitivity conjecture instantly followed from this result.

When Huang's paper landed in Mathieu's inbox, her first reaction was "uh-oh," she said. "When a problem has been around 30 years and everybody has heard about it, probably the proof is either very long and tedious and complicated, or it's very deep." She opened the paper expecting to understand nothing.

But the proof was simple enough for Mathieu and many other researchers to digest in one sitting. "I expect that this fall it will be taught – in a single lecture – in every master's-level combinatorics course," she messaged over Skype.

Huang's result is even stronger than necessary to prove the sensitivity conjecture, and this power should yield new insights about complexity measures. "It adds to our toolkit for maybe trying to answer other questions in the analysis of Boolean functions," Servedio said.

Most importantly, though, Huang's result lays to rest nagging worries about whether sensitivity might be some strange outlier in the world of complexity measures, Servedio said. "I think a lot of people slept easier that night, after hearing about this."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